

AD-MaaS Business Logic

Seneschal: <https://seneschal.base44.app> ... Demand-generation implementation example
Federation: <https://federation.base44.app> ... Urban-operation packaging study framework

The concepts presented in this portfolio are not proposals for implementation. They demonstrate my concrete understanding of the strategy and execution requirements needed to make autonomous-mobility businesses commercially viable.

The success of AD-MaaS is determined not by autonomous driving technology itself, but by a business structure that converts operation into profit. As driver labor cost decreases through automation, the dominant cost factor converges toward energy cost, while vehicle utilization is required to recover CAPEX and load factor is required to increase revenue per trip. Seneschal smooths demand fluctuations by matching customer availability with mobility and services, while motion-sickness-free vehicle control transforms travel time into work, rest, and service value.

Because vehicle-grade quality requirements are strict, next-generation vehicle technologies should first be developed for stationary applications, then installed into service vehicles as they become ready. Service vehicles can absorb CAPEX through high utilization and high load factor, enabling a DevOps-style development model unlike conventional mass-production vehicles. Federation increases vehicle utilization through time-based service exchange and ridesharing, and increases load factor through passenger-freight integration and collaborative logistics, while planned mobility schedules optimize PPA energy management and enable AVP-integrated fueling to establish a carbon-neutral energy platform for autonomous mobility. AD-MaaS is an urban operating system that controls demand, vehicles, energy, and infrastructure as one system to convert utilization, load factor, and energy cost into profit.

PPA Service for Autonomous Mobility Foundation

As autonomous driving reduces driver labor cost, energy becomes the dominant operating-cost factor. The PPA service therefore establishes structural energy-cost competitiveness for AD-MaaS through distributed energy infrastructure.

Corporate PPA services support Scope 3 decarbonization while establishing distributed eVTOL operating hubs. Idle corporate vehicles are shared for employee commuting to improve fleet utilization and reduce Scope 3 transportation emissions. Residential trigeneration services establish distributed neighborhood mobility hubs, providing the node density required for daily autonomous mobility services.

Blockchain-enabled P2P energy and mobility sharing connects corporate and residential nodes into a shared mobility and energy ecosystem, while renewable electricity, water-pipeline micro-hydropower, magnetic refrigeration, hydrogen production, superconducting power cables, liquid-hydrogen pipelines, and liquid-hydrogen-cooled HPC establish the infrastructure foundation for infrastructure-coordinated autonomy and continuous ODD expansion.

Infrastructure Layer	Control Variable	Target	Business Purpose
Renewable PPA	Renewable supply ratio	Maximize	Low-cost renewable electricity
Water Infrastructure	Micro-hydropower output	Maximize	Distributed baseload generation
Hydrogen Production	Electrolysis utilization	Maximize	Hydrogen production
Cooling	Magnetic refrigeration efficiency	Maximize	LH ₂ production & HPC cooling
Energy Network	Superconducting cable utilization	≥80%	Inter-node power sharing
Hydrogen Network	LH ₂ pipeline utilization	≥70%	Hydrogen distribution
HPC Node	AI compute availability	≥99.9%	Infrastructure AI computing
Infrastructure Coordination	Infrastructure coverage	Expand	Vehicle–infrastructure cooperation
Autonomous Mobility	ODD coverage	Expand	City-wide autonomous operation

AVP-to-MaaS Execution

This concept begins with an existing OEM customer use case: enabling refueling to be completed while the user is shopping. → A: time control
By integrating infrastructure-coordinated automated valet parking (AVP) with fueling, vehicles autonomously park, refuel, and return within dwell time, eliminating dedicated refueling stops. → B: time + location control
As autonomous driving progresses, parking demand shifts, creating underutilized parking spaces repurposed as operational nodes. By leveraging these assets, dual-use infrastructure enables daytime refueling for private users and nighttime fleet operations. → C: time allocation control
Liquid-hydrogen-cooled HPC nodes provide compute capacity for infrastructure-coordinated autonomy. Any ODD expansion remains subject to scenario coverage, V&V evidence, a validated safety case, and regulatory approval. Applied to large-scale parking operators, operational nodes exchange modular service containers according to time-of-day demand, balancing utilization across mobility, logistics, retail, and public services. This transforms Node into a MaaS foundation that mitigates demand fluctuations and increases asset utilization. → D: operational control
Extending this logic to public roads evolves the system into a city-wide mobility platform. → E: flow control
Through autonomous driving, vehicles synchronize in time and space, forming organized carrier operations rather than independent movements. → F/G: synchronization / platoon control
This expansion to public roads and coordinated carrier systems enables a new transportation system. Because this system is built on coordinated flows and shared infrastructure, Autonomous mobility expands from vehicle operation into integrated mobility and infrastructure coordination. → H: system-level control

Logic	Layer	KPI	Formula	Value	Unit
A	Demand	Demand generation rate	$\text{generated demand} \div \text{total potential demand}$	≥0.30	ratio
A	Demand	Conversion rate	$\text{booking} \div \text{request}$	≥0.23	ratio
B	Demand	Matching rate	$\text{matched} \div \text{request}$	≥0.90	ratio
D	Demand	Trip request rate	$\text{executed trips} \div \text{matched}$	≥0.95	ratio
D	Operation	Dispatch success rate	$\text{completed dispatch} \div \text{dispatch request}$	≥0.95	ratio
D	Operation	Dispatch response time	$\text{total dispatch response time} \div \text{dispatches}$	≤60	sec
D	Operation	Utilization	$\text{operating time} \div \text{total time}$	≥0.75	ratio
D	Operation	Fleet idle ratio	$\text{idle vehicles} \div \text{total vehicles}$	≤0.25	ratio
F	Vehicle	Latency	response time	≤1	sec
F	Vehicle	Control delay	actuation delay	≤20	ms
F	Vehicle	Planning latency	planning delay	≤100	ms
G	Vehicle	Control stability	spacing variance	≤5	%
E	Infrastructure	Throughput	$\text{vehicles} \div \text{time}$	≥40	veh/h
C	Infrastructure	Node utilization	$\text{used nodes} \div \text{total nodes}$	≥0.80	ratio
F	Infrastructure	Node latency	node delay	≤50	ms
H	Infrastructure	Energy cost per trip	$\text{energy cost} \div \text{trips}$	≤¥500	JPY/trip

Business Development for Autonomous Mobility Execution

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Executive Summary: Mobility Control

Business architect with 29 years of experience in automotive, semiconductor, and energy industries, specializing in structuring and executing mobility-related businesses under real operational and financial constraints. I have led large-scale business structuring and execution while coordinating a business organization of approximately 100 members and a development organization of approximately 500 engineers, with annual budgets of around ¥100B, and have consistently delivered structured decision frameworks to executive leadership on a daily and weekly basis.

At Honda, in next-generation business development, I formulate GTM strategies for Go/No-Go assessment and support investment decision-making. As FCAD-MaaS LPL, I drive integrated product and business strategy across autonomous-driving operations platforms, MaaS vehicles, and service hubs. Separately from my Honda responsibilities, I developed Seneschal as a customer application, used Federation to examine fleet-operations consoles, and built an independent portfolio model connecting demand and fleet operations to KPIs and P&L.

Business Execution Responsibility

I have led autonomous mobility business development across engineering, operations, service planning, and executive decision-making under real operational constraints, taking responsibility for structuring autonomous mobility business conditions into executable operations.

- Structured business conditions across vehicle, energy, and service domains under real operational and financial constraints.
- Delivered decision frameworks directly to executive leadership and enabled informed decision-making
- Translated strategic directions into executable plans and drove cross-functional alignment
- Connected vehicle, operation, energy, and service requirements with business feasibility, ensuring consistency between system design and economic outcomes

In practice, this involved working across multiple layers of stakeholders and constraints.

- Coordinated with engineering teams (~500 engineers) to align technical feasibility with business conditions
- Engaged with business planning and corporate strategy functions to structure investment and revenue logic
- Interfaced with executive leadership through regular reporting cycles, incorporating feedback into evolving decision frameworks
- Managed conflicting priorities between short-term execution and mid- to long-term strategic direction

This role required not only structuring analysis, but also driving convergence of stakeholders with different objectives, and translating complexity into executable actions.

Business Impact

- Contributed to reassessment of next-generation powertrain strategy at corporate level through structured analysis and decision framework
- Played a key role in initiating a Honda-branded commercial-truck program under the current four-wheel business structure by defining business and system conditions
- Contributed to restructuring of hydrogen business and energy strategy through integration of business and technical perspectives
- Structured a mobility business model under real operational constraints and enabled its evaluation at executive level
- Developed GTM and investment decision materials supporting Go/No-Go assessments

Business Architecture Capability

I specialize in structuring mobility businesses by integrating multiple domains into a single executable framework.

- Demand conditions are defined based on usage patterns and translated into measurable utilization and service requirements
- Operations are structured as continuous systems covering dispatch, routing, and control, integrated with logistics and energy
- Vehicle requirements are defined in relation to utilization, cost, and operational continuity under business constraints.
- Infrastructure and energy systems are structured to support scalability, with clear linkage between supply, utilization, and cost

Through this approach, I connect KPI, utilization, cost, and revenue into a consistent structure, enabling complex systems to be evaluated and executed as business.

KPI to P&L Capability

- Structuring KPI → utilization → cost → revenue into one integrated business model
- Integrating demand, operations, vehicle, and infrastructure into a unified decision framework
- Translating complex technical systems into executable business structures
- Driving alignment across engineering, business, and executive layers

Positioning: Autonomous Mobility Business Development

Autonomous mobility requires integration of multiple domains into a single executable system.

I structure and execute autonomous mobility business operation under real-world constraints.

Product Execution Case (Honda FCAD-MaaS)

At Honda, as FCAD-MaaS LPL, I drive integrated product and business strategy across autonomous-driving operations platforms, MaaS vehicles, and service hubs, while formulating GTM strategies and supporting Go/No-Go and investment decisions. Separately from my Honda responsibilities, the accompanying independent portfolio connects demand and fleet operations to KPIs, cost structure, P&L, and investment logic.

1. Demand & Experience

Examined how user intent, autonomous-driving capabilities, and service conditions could be connected within an executable system concept.

- Structured relationships among user intent, ODD / DDT, and service feasibility
- Evaluated system configurations incorporating external autonomous-driving technologies and services

Execution:

Worked across business planning, service design, and system teams to evaluate how user requirements could be translated into system conditions and service-level utilization.

Result:

Evaluated an autonomous-driving system configuration incorporating Tier IV and SoftBank technologies and services, linking user intent with ODD / DDT feasibility and service-level utilization.

2. Service Operations & Orchestration

Examined FCAD-MaaS as an integrated system connecting autonomous driving, logistics DX, and energy management.

- Defined system architecture linking autonomous transport, logistics, and hydrogen supply
- Examined interfaces and handoff conditions across component systems

Execution:

Worked with multiple logistics-DX vendors and internal stakeholders to define system requirements and architecture for AGV-based hydrogen replenishment.

Result:

Defined an architecture integrating AGV-based hydrogen replenishment with a logistics-DX-connected platform to reduce vehicle idle time and support continuous autonomous-mobility services.

3. Vehicles & Embedded Intelligence

Defined vehicle architecture as a business-critical component linking utilization, energy efficiency, and operational continuity.

- Structured vehicle requirements integrating powertrain, control, and operational conditions
- Evaluated vehicle architecture under business constraints, including utilization and cost

Execution:

Worked with vehicle development and powertrain teams to align technical architecture with business conditions, structuring decision frameworks used to evaluate next-generation vehicle direction.

Result:

Defined vehicle architecture required for autonomous mobility, combining flat FC skateboard and swap body to improve utilization, and HTS-based control responsiveness to prevent motion sickness and ensure stable autonomous driving.

4. Energy & Urban Infrastructure

Defined infrastructure and energy systems as the foundation of scalability and cost structure in autonomous MaaS.

- Structured integrated energy systems combining hydrogen and electricity
- Defined infrastructure deployment model aligned with mobility operation

Execution:

Coordinated across energy, infrastructure, and business teams to structure feasibility conditions, integrating production, distribution, and utilization into a consistent business framework.

Result:

Structured a stationary energy-system concept using mill-based hydrogen production and hydride-ion conductor storage alloy, and proposed infrastructure-first deployment to address vehicle-integration constraints, reduce fuel cost, and evaluate a cost-competitive MaaS foundation for autonomous mobility.

Business and System Integration

Integrated autonomous-driving system, vehicle, energy, and infrastructure requirements and supported their evaluation under real-world constraints.

- Coordinated cross-functional stakeholders to align technical and business conditions
- Supported Go/No-Go and investment decisions through structured GTM, business, and system analysis
- Drove alignment across engineering (~500), business planning, and executive layers by iteratively refining business and system structures, ensuring feasibility and decision readiness

Result:

Structured autonomous mobility as an evaluable business and system concept and enabled investment-level assessment.

Separately, through independent portfolio development, I developed Seneschal as a customer application, used Federation to examine fleet-operations consoles, and connected the resulting system concepts to a KPI-to-P&L model.

Positioning: Autonomous Mobility Business Development

This work demonstrates the ability to structure and execute autonomous mobility as a business by integrating multiple domains under real constraints and driving alignment across organizations.

Reusable Product Architecture

The following experiences substantiate the four-layer mobility business structure established at Honda and form the foundation for designing and executing scalable autonomous mobility business operation.

Professional Experience

Toyota Connected : Mobility Service Planning (2017–2022)
Renesas Electronics : Business Development Manager (2022)
Sharp : Global Business Development Manager (2014–2016)
Denso SI / Samsung SDI : Project Leadership / Business Development (2013–2014)
Denso : Sales, Planning, Program Management (1997–2012)

Foundation of executable mobility structure

The following experience established a consistent capability to structure mobility-related business across demand, operations, vehicle, and infrastructure domains prior to Honda, by translating fragmented technical, operational, and organizational elements into executable systems.

1. Demand & experience

Defined how user behavior, service conditions, and system constraints are translated into demand, utilization, and executable service requirements.

- Toyota Connected: Structured mobility service concepts linking vehicle behavior, infrastructure conditions, and service availability, including autonomous mobility planning and infrastructure-linked AVP service design, defining how user-side requirements connect to dispatch and service execution
- Sharp: Built demand models based on connected services integrating devices, sensors, and user interaction, converting behavioral triggers into service demand and monetizable opportunities
- Denso SI: Designed service concepts linking vehicle state, infrastructure coordination, authentication, and payment, structuring how user action, infrastructure condition, and execution workflow are connected
- Renesas: Connected user-level requirements with AD/ADAS development and validation environments by linking UX-side expectations with simulation and model-based development conditions

Result: Established capability to translate user behavior and service conditions into executable demand structures rather than abstract assumptions.

2. Service operations & orchestration

Defined how mobility services are executed through coordination of vehicles, infrastructure, and systems under real constraints.

- Toyota Connected: Structured service execution linking demand, dispatch, fleet coordination, and infrastructure-linked operation, defining mobility as a service system rather than a function
- Denso SI: Built service models integrating mobility, energy, payment, and infrastructure systems, structuring operational flow across charging, authentication, billing, and infrastructure coordination
- Denso: Coordinated OEM programs aligning engineering, cost, delivery, and customer requirements across stakeholders, connecting planning assumptions with rollout execution
- Renesas: Structured development processes integrating simulation, validation, and system design, enabling AD/ADAS and E/E architecture development as an executable system

Result: Established capability to structure mobility operation as a coordinated execution system under technical and organizational constraints.

3. Vehicles & embedded intelligence

Defined how vehicle systems are structured as integrated components within mobility service and business architecture.

- Denso: Coordinated vehicle system integration across OEM programs including ADAS-related development, aligning specification, cost, and system requirements
- Samsung SDI: Developed battery and energy systems supporting electrified vehicle architecture, linking vehicle-side requirements with infrastructure-side energy usage
- Sharp: Integrated sensing and computing technologies into vehicle-related systems including NVIDIA-based platforms and connected sensing architecture
- Renesas: Structured system-level development linking semiconductor, control, and vehicle architecture, connecting AD/ADAS requirements with implementation feasibility

Result: Established capability to position vehicle systems as determinants of utilization, cost, and service execution within mobility systems.

4. Energy & urban infrastructure

Defined how infrastructure and energy systems support mobility operation, economics, and scalability.

- Toyota Connected: Structured infrastructure-linked mobility systems integrating parking, charging, and vehicle-infrastructure coordination
- Denso SI: Developed infrastructure systems linking mobility, energy, authentication, and financial services, demonstrating service-integrated infrastructure models
- Samsung SDI: Built energy storage and utilization models supporting infrastructure deployment and electrified mobility
- Sharp: Designed smart city systems integrating infrastructure, devices, and mobility services at urban scale

Result: Established capability to structure infrastructure and energy systems as business components determining cost structure and scalability.

Integrated execution capability

Across these experiences, I consistently translated fragmented technologies, organizations, and operational constraints into executable business structures, aligning technical feasibility with business viability. This foundation enabled these capabilities to be applied at Honda through autonomous-driving system and business planning, demonstrating consistency and reproducibility across different domains and organizations.

Execution Model Under Constraints

I have consistently delivered results by structuring fragmented technical, operational, and business constraints into a single executable system.

Across multiple roles and domains, I have translated complex conditions into decision frameworks, aligned stakeholders, and driven execution under real-world constraints. This approach is based on defining structure, connecting elements, and making them actionable.

At Honda, I apply this capability to product and business strategy integrating autonomous-driving operations platforms, MaaS vehicles, and service hubs.

This demonstrates a repeatable capability to structure and execute complex systems as business.

How I Execute

My approach is to define a consistent structure that connects technical feasibility, operational conditions, and business viability, and then align stakeholders across engineering, business, and management.

Through iterative refinement, I translate concepts into executable systems, ensuring that all elements are connected within a single decision framework and can operate under real-world constraints.

Repeatable Business Execution

Autonomous-mobility businesses require an end-to-end leader who can convert mobility needs into product strategy, business models, and executable launch plans

- I will lead the full business cycle—from customer value and use-case definition through revenue model, business scheme, product requirements, schedule, budget, organization, alliance and investee governance, sales and field implementation
- I will align customers, operators, alliance partners and internal functions through clear ownership, milestones, decision gates and KPIs, while translating market and operational evidence into proposals for vehicle, autonomous-driving, infrastructure and service development
- My focus is to establish safe, repeatable and commercially viable operations, validate value and economics in the field, and scale proven services across use cases and regions without compromising quality
- This will advance organization-wide mobility transformation, safer transportation and a sustainable future revenue pillar.

Full-Stack Mobility Integration Across Domains, Roles and Constraints

Context (Period / Role)	Achievement	Outcome
1997–2003 / Product planner / regulatory / Industrial control	Built agricultural HVAC layout & construction D-EMS products	Achieved 100% market share & navigated regulations (DENSO)
2001–2003 / Sales strategist / Organizational Fragmentation	Involved in JV of JTEKT / ADVICS & entered NISSAN ecosystem	Built new biz channel & scaled ADAS capital intensive (DENSO)
2004–2006 / Powertrain sales / cost-constraints. / electricity	Defended DAIHATSU business & entered China FAW ecosystem	Solved cost constraints. & acquired AISAN with By-Wire tech (DENSO)
2007–2009 / Body-parts sales / multi-stakeholders / ADAS	Promoted MAZDA ADAS & E/E architecture & entered china FAW	Achieved China OEM entry & aligned multi-stakeholders (DENSO)
2010–2012 / Strategy manager / TOYOTA global emerging market	Led TOYOTA strategy internal & India, Brazil, Egypt projects	Drove TNGA program & scaled global OEM operations (DENSO)
2013 / Battery key account manager / BESS infrastructure	Signed joint Dev agreement & managed automotive promotion	Led all SAMSUNG automotive business Develop & exhibition (SAMSUNG SDI)
2013–2014 / System engineer / BEV infrastructure / Fintech	Signed agreement of cloud native EV charging with QR billing	Demonstrate with Fukuoka local government & EXPO (DENSO SI)
2013–2014 / Service planner / microgrid / MaaS operation	Contributed NEDO Project of Microgrid by Multimodal Service	Selected Anjo demo & validated EV grid integration tech (DENSO SI)
2014–2016 / IoT Strategic sales / GAFAM / ecosystem Integration	Built Amazon-linked demand model & won service prototype	Developed Amazon Dash replenishment service (SHARP)
2014–2016 / Autonomous biz-dev / LIDAR & automobile PF	Entered NVIDIA-based platform mass production & GOOGLE LIDAR proto	Overcame technology immaturity for automobile quality (SHARP)
2014–2016 / Smart City planner / V2I·V2P urban deployment.	Installed Communication Device to CISCO for NEDO Smart City Demo	Demonstrated with Fukuyama Local Government (SHARP)
2014–2016 / Automobile biz-dev / investment constraints.	Designed mirror centric PV roof & transparent display module	Initiated automobile company starting from camera (SHARP)
2017–2022 / Service & business planning / Connected car service	Launched TOYOTA connected car platform service business	Established mobility foundation (TOYOTA Connected)
2017–2022 / New business development / MaaS application	Built Application to maximize utilization of MaaS Business	Concretized TOYOTA 20XX Concept (TOYOTA Connected)
2017–2022 / System architect / V2I AVP / MaaS operation	Built Business Model as a core foundation of MaaS Business	Built Scalable Urban Service Model (TOYOTA Connected)
2022 / Platform strategy / AD EV / semi-con constraints.	Developed E/E simulation & HILS MBD development environment	Acquired ALTIUM & enabled SDM / MBDS business (RENESAS)
2022–Now / Hydrogen powertrain strategy / Energy transition	Structured hydrogen & MaaS business conditions	Drove reassessment of Electrification Investment Decision (HONDA)
2022–Now / Vehicle architect / MaaS vehicle / Logistic systems	Led company decision to begin development of HONDA's first truck	Drive Reassessment to MaaS Vehicle Development (HONDA)
2022–Now / Urban architect / transition investment constraints.	Led company-wide structural transformation through ecosystem	Enabled the SDM organization to restart MaaS development. (HONDA)

Proven across all layers of mobility systems

Autonomous Mobility Business Execution Architecture

— Driving autonomous mobility business development through AD/ADAS system foundation —

Autonomous Mobility Business Definition

This architecture is defined as a control system where KPI, system behavior, and business outcomes are directly connected. This architecture directly connects KPI, system behavior, and business outcomes as a single control system. The objective is to build an AI-enabled autonomous mobility business architecture that improves utilization, load factor, and unit economics.

This is achieved by integrating mobility, logistics, energy, and urban services through a common operating architecture, while structurally improving:

- utilization
- load factor
- unit economics

This proposal defines a business architecture that makes this system executable, economically grounded, and deployable.

Autonomous Mobility Business Execution

This proposal defines a phased development and deployment plan for autonomous mobility business execution. Today's mobility systems fail to scale because demand, fleet, energy, and operations are still managed separately. This architecture integrates these elements into a unified autonomous mobility business structure, using autonomy as the foundation and AI as the means to operate autonomous mobility at business scale. System performance directly translates into revenue through utilization, load factor, and energy efficiency:

- Demand creation synchronized with dispatch
- Utilization maximization through modular vehicle operation
- Load factor optimization through dynamic vehicle composition
- Energy cost reduction through on-site hydrogen production
- Real-time operational control through distributed node coordination and AI-enabled optimization

Autonomous Mobility Business Architecture

Objective: To define an executable autonomous mobility business structure where AI-driven control logic is applied through autonomous driving to integrate demand, operations, vehicles, and infrastructure.

This architecture shifts mobility from fragmented services into a unified operating system where utilization, load factor, and energy efficiency are structurally optimized.

AI is applied as controlled logic across four layers.

1. Demand Orchestration

2030: Coordinated AI Assistant

A calendar-based interface connects user schedules with service availability, enabling demand generation through matching and reservation-linked mobility.

2035: Trusted AI Partner

A 4D digital platform integrated with map infrastructure synchronizes city-state and user behavior in real time, enabling continuous demand generation and capture to maximize load factor.

2. Operation Control

2030: Integrated Control

Energy management and dispatch are coordinated through decision logic that can later be augmented by AI, enabling stable and efficient operations under constrained conditions.

2035: Synchronized Control

Real-time synchronization of in-car and out-car hydrogen and electricity flows creates the control foundation for AI-assisted optimization, while structurally improving business viability.

3. Fleet Vehicle

2030: Swap Carrier

A modular vehicle structure enables continuous operation through container exchange, maximizing utilization.

2035: Composed Carrier

Vehicles dynamically couple and reconfigure during operation, enabling real-time redistribution of passengers and cargo to maximize load factor.

4. Distributed Infrastructure

2030: Urban Node

Distributed hubs provide localized energy and compute support, enabling operational independence and cost efficiency.

2035: Node Mesh

Distributed infrastructure is synchronized as a network, enabling real-time coordination between vehicles, energy, and computation.

This architecture enables a system where mobility, energy, computing, and AI-based decision-making operate as a unified layer, enabling scalable autonomous mobility business operation.

AI-Enabled Urban Mobility Roadmap

Autonomous mobility becomes deployable only when a production-ready driving stack, closed-loop learning, auditable safety, fleet operations and service economics are developed as one system. This roadmap first establishes that industrial baseline, then extends AI control from vehicles to distributed infrastructure and city-scale operations.

TIER IV Technology Foundation & Capability Build-Up

Capability	TIER IV Technology Foundation	Roadmap Focus
Product / Release Governance	Web.Auto — cloud-native DevOps platform for development, testing, deployment and operational lifecycle management.	Stage deployment through safety, service-readiness, availability, recovery-time and unit-economics gates.
Driver / Sensor Platform	Pilot.Auto + Edge.Auto — scalable driving software with vehicle/sensor platform integration across multiple vehicle classes.	Common driving stack, hardware abstraction, calibration, fail-operational design and rapid vehicle onboarding.
Data / Active Learning	Co-MLOps / Web.Auto — fleet data collection, anonymization, annotation, rare-event mining and continuous learning.	Closed loop: detect → anonymize → select → label → train → validate → staged release.
Simulation / V&V	L4 V&V + Web.Auto — scenario-based verification and validation using simulation, replay, closed-course and public-road evidence.	ODD-specific scenario coverage, acceptance criteria, critical-event recall, safety evidence and validation gates.
Safety / Remote Support	Pilot.Auto safety functions — ODD monitoring, fallback/MRM, remote monitoring and assistance with clear authority boundaries.	Claims, evidence, escalation, incident response, cybersecurity, rollback, audit trail and remote-support limits.
Fleet / Service Operations	Web.Auto / FMS — dispatch, fleet monitoring, maintenance, incident response and service operations integrated with Autoware.	Connect the driving stack with fleet and service control; manage SLA, availability, intervention, energy and cost/hour.
Scale / Economics	Autoware + Pilot.Auto + Web.Auto + Edge.Auto — reusable, configurable platform for multi-vehicle deployment and ODD expansion.	Track utilization, launch lead time, ODD rollout speed and lifecycle revenue; separate common and use-case-specific functions.

The industrial baseline combines a production-ready driver, closed-loop learning, scenario-based V&V, auditable safety, remote support, fleet operations and repeatable ODD deployment. The roadmap establishes these capabilities first, then differentiates through city data, infrastructure coordination, demand orchestration and business optimization.

AI-Controlled Mobility System

This section defines how AI functions as control logic across layers and is executed through autonomous driving systems. AI is applied differently at each layer — from demand prediction to system-wide coordination — but all functions are integrated through a unified control structure that enables real-time execution at city scale.

Layer	Cloud	Distributed Infrastructure	Fleet Vehicle
Perception	2030: Env-state estimation (fusion, uncertainty propagation) / 2035: City-wide world-state estimation (global sync, uncertainty minimization)	2030: Roadside perception augmentation (edge fusion, low-latency inference) / 2035: Cooperative perception (shared state, spatiotemporal fusion)	2030: Onboard perception (LiDAR / Camera / Radar, ≤10ms) / 2035: Infra-coordinated perception (shared scene state)
Planning	Dispatch optimization (forecast, constrained search) / 2035: City mission optimization (global scheduling, real-time recompute)	2030: Zone operation optimization (routing / berth / node) / 2035: Real-time replanning (node sync, global constraints)	2030: Local path / motion planning / 2035: Cooperative trajectory planning
Control	2030: Learning-based control (constraints, stability) / 2035: Global closed-loop optimization (adaptive)	2030: Edge feedback control (low-latency, constraints) / 2035: Distributed synchronized control (coordination, load)	2030: Motion control (MPC, ≤50ms, constraint-aware) / 2035: High-frequency adaptive control (≤10ms, cooperative)
Coordination	2030: Fleet-state coordination (dispatch-state sync) / 2035: Fleet-wide coordination (global sync, decision integration)	2030: Node-state coordination (local sync) / 2035: Node Mesh synchronization	2030: Vehicle-level coordination / 2035: Platoon coordination

ODD / DDT as Revenue Driver

ODD / DDT transition defines idle time and directly determines utilization and revenue structure.
DDT Loss = (1-TSR) x Recovery time + TOT, Revenue (OAR x UAC) x (1 ÷ (Trip+(1-TSR) x Recovery + TOT) x FLC x Load

Execution Sequence & Exit Gates

The roadmap advances only when technical evidence, operational readiness and unit economics are verified together.

Phase	Capability Build	Exit Gate	Business Outcome
1. Industrialize	Common driving and sensor stack; closed-loop data; scenario V&V; auditable safety case; remote operations and field support.	Repeatable launch within the initial ODD, with controlled fallback, rollback, incident response and configuration management.	Production service with measurable safety, availability, intervention rate, recovery time and cost/hour.
2. Federate	Fleet orchestration; RSU / OBU coordination; shared data and simulation; distributed infrastructure and service control.	Cross-vehicle and cross-site operation under common interfaces, authority boundaries, release procedures and KPI governance.	Lower intervention, recovery time and idle capacity; faster vehicle, site and ODD onboarding.
3. Scale	4D demand orchestration; Node Mesh; energy-compute integration; evidence-led quantum computing, sensing and optimization.	Multi-ODD, multi-service and multi-city performance with resilience, traceability and viable unit economics.	Scalable urban mobility platform with recurring service, infrastructure and lifecycle revenue.

Urban AI Operating Architecture

A TIER IV-based capability progression from Autoware / Pilot.Auto and Edge.Auto to Web.Auto, Co-MLOps, L4 V&V and city-scale operations.

1. UNDERSTAND & PROTECT	2. CONNECT & IMPROVE	3. OPERATE & SCALE
Real-world intelligence and coordinated safety	Governed data, validation and continuous learning	Distributed execution and city-scale orchestration
Autoware Perception — Builds a multimodal understanding of people, vehicles, infrastructure, road conditions and the surrounding urban environment.	Edge.Auto — Connects sensors, vehicle compute and edge functions through standardized interfaces for autonomous-driving deployment.	4D demand orchestration — Predicts where, when and why demand will emerge across mobility, logistics, energy and public services.
Autoware Prediction / Planning — Predicts human and vehicle intent, interaction patterns, anomalies and uncertainty to support context-aware decisions.	Co-MLOps — Federates fleet data collection, annotation, learning and validation without forcing every participant into a single centralized system.	Rare-event learning — Prioritizes edge cases, near misses and low-frequency, high-impact events for targeted learning and validation.
Pilot.Auto — Synchronizes autonomous-driving functions, shared awareness and timely intervention across the vehicle and surrounding environment.	Web.Auto DevOps — Converts operational evidence into simulation, validation, controlled deployment and measurable continuous improvement.	RSU / OBU coordination — Distributes perception, routing, safety and operational tasks across roadside and onboard systems.
Pilot.Auto Safety / MRM — Integrates detection, prediction, risk assessment and intervention, including fallback and minimum-risk maneuvers.	L4 V&V — Tests ODD changes, scenarios, service patterns, traffic conditions and rare events before field deployment.	Node Mesh synchronization — Coordinates vehicles, roadside nodes and local service hubs as a resilient, distributed operating network.
Multimodal real-world understanding — Fuses cameras, sensors, maps, V2X signals and contextual data into a consistent operational picture.	Web.Auto Operations — Captures outcomes, deviations and system performance to update models, rules and operating procedures.	Energy-compute integration — Allocates charging, storage, edge computing and fleet tasks against demand, cost, capacity and resilience.
Predictive risk and uncertainty management — Identifies emerging conflicts early enough to preserve human decision time and maintain a safe operating margin.	Web.Auto Simulation / CI/CD — Links design assumptions, simulation results, field evidence and software updates throughout the lifecycle.	Quantum computing and sensing — Explores optimization and sensing gains for high-dimensional mobility, infrastructure and urban-service operations.
Coordinated safety — Aligns human handover, vehicle fallback, infrastructure assistance and accountable decision-making across the operating environment.	Web.Auto Release Governance — Releases only validated updates with traceability, approval, rollback capability and measurable safety evidence.	Quantum-enhanced perception — Applies advanced sensing and inference only where they demonstrate meaningful gains over classical approaches.
Shared scene awareness — Makes relevant hazards, intent and operating context understandable across people, vehicles, infrastructure and service operators.	City-data integration — Connects mobility, energy, logistics and public-service operations through governed, reusable data products.	Full-city synchronization — Coordinates cross-domain actions while preserving local autonomy, safety boundaries and service continuity.
Safety-to-ODD expansion — Extends system capability only after evidence demonstrates acceptable risk and operational readiness in each target condition.	Autoware / Web.Auto closed-loop execution — Deploys approved models, policies and operating rules back to vehicles, RSUs, infrastructure and service operations.	Autonomous city-state optimization — Continuously balances safety, service quality, capacity, energy consumption, resilience and accessibility.
Human trust and adoption — Explains system state, preserves appropriate human authority and captures acceptance signals from users and operators.	Governance, KPI and business control — Monitors safety, availability, unit economics, energy, inclusion, service quality and lifecycle performance.	Scalable business execution — Converts technical gains into repeatable operations, partnerships, asset management and sustainable lifecycle value.

Cross-Layer Decision Rights & Release Gates

Control Domain	Primary Authority	Evidence Returned Upward	Release / Deployment Gate
Vehicle / Edge	Immediate perception, motion control, fallback, minimum-risk manoeuvre and local safety intervention.	Uncertainty, hazard traces, interventions, sensor and actuator health, and safe-state results.	Latency, hazard-response coverage, fail-operational capability and safe-state completion.
Fleet / Infrastructure	Dispatch, routing, RSU / OBU coordination, remote assistance, node control and energy-compute allocation.	Availability, dwell and recovery time, node congestion, service exceptions, energy use and capacity.	SLA, intervention rate, recovery time, operational continuity and ODD-support readiness.
Cloud / City	Model and policy release, scenario validation, governance, cross-domain learning and city-scale optimization.	Scenario coverage, causal evidence, audit trail, service quality, cost and lifecycle-performance KPIs.	Traceability, rollback readiness, validated KPI improvement, resilience and sustainable unit economics.

Validated policies, models and operating rules move downward; evidence, uncertainty, incidents and business KPIs move upward. Safety authority remains local, while learning and optimization remain globally governed, measurable and auditable.

Across these three layers, the operating cycle remains Observe → Understand → Learn → Evaluate → Execute → Observe Again. Real-world signals are interpreted and protected at the edge; governed data is federated and converted into validated improvements; approved policies and models are then deployed back to vehicles, RSUs, infrastructure and urban operations. Evidence and learning move upward, while only validated policies and control updates move downward. The architecture scales from coordinated assistance to cross-domain orchestration across mobility, logistics, energy and public services, while preserving human accountability, fallback and rollback. Quantum computing, sensing and quantum-enhanced perception are introduced as evidence-led accelerators—not assumptions—and must demonstrate measurable gains in safety, resilience, service quality, capacity or lifecycle economics before deployment. The result is a practical path from isolated AI functions to a trusted, continuously improving urban operating system.

Satofumi Goto

Layer-integrated Control & Business Architecture

This architecture defines how business requirements are translated into executable system design through AI-driven control and autonomous driving. It connects KPI targets with technical execution, ensuring that system performance directly drives economic outcomes. Executable mobility operation This section maps business KPIs to control logic and AI-relevant technical conditions required to achieve them.

2030			2035		
Logic		Input	Target	Logic	Input
Perception	Match ≤500ms [A]	Cal ±5min	CVR ≥23%	Match ≤100ms [A]	Behavior ±1s
	Update ≤5min	Res ≤1s	Wait ≤6min	Update ≤1s	History ≤10s
	Trigger ≤1s	Avail ≤500ms	Cancel ≤12%	Trig ≤100ms	Avail ≤100ms
	Conv ≤1min		Eff conv≥19.6%	Conv ≤10s	
	Err ≤20%		Match ≤500ms	Err ≤5%	
Planning	Succ ≥85%		Err ≤20%	Reflect ≥95%	
			Trigger ≤1s		
	Disp ≤1min [B]	Fleet ±50m	Trips ≥73/d	Disp ≤10s [B]	Fleet ±5m
	Route ≤5s	ETA err ≤3min	Loaded ≥14/d	Route ≤1s	Traffic ±1s
	Alloc ≤10min	Traffic ≤30s	Resp ≤8min	Alloc ≤1min	Sync ≤5s
Control	Loop ≤5min		ETA err ≤3min	Loop ≤1min	
	Succ ≥95%		Empty ≤25%	E-int ≥95%	
			Util ≥75%	Succ ≥99%	
			Route ≤5s		
	MPC ≤50ms [C]	Percep ≥95%	Price ≥¥2,200	MPC ≤10ms [C]	Perce ≥99.9%
oordination	Traj ≤50ms	Plan ≤200ms	Ctrl ≤50ms	Traj ≤10ms	Plan ≤50ms
	Conv ≤100ms	V2X ≤20ms	Plan ≤200ms	Conv ≤10ms	Sync ≤10ms
	Stability ±15%		Err ≤30cm	Stability ±5%	
	Fail-safe ≤100ms		Intv ≤1/100km	Coop ≥95%	
			Util ≥87.5%	Fail-safe ≤10ms	
	Alloc ≤1min	Energy ±5% [D]	OPEX bound	Alloc ≤100ms	Energy ±1% [D]
	Compute ≤1s	Comp ≤100ms	Comp ≤150ms	Comp ≤10ms	Comp ≤50ms
	Refuel ≤5min	Sync ≤1min	Eff ≥75%	Refuel ≤1min	Sync ≤10s
	Conv ≤10min		Util ≥78%	Strategy ≤1h	
	S-D ≥0.78		Eff energy ≥58.5%	Conv ≤1min	
	Load err ≤10%		Refuel ≤2/d	S-D ≥0.96	
			Var ≤±5%	Load err ≤3%	

Operating index (dimensionless) = Utilization × Load Factor × Energy Efficiency; profit still requires price, volume, and cost. Available service time = scheduled time – [(1–TSR)×recovery time + TOT]; Revenue = Trips × Seats × Load × Fare.

Layer-Integrated Business Architecture

This section defines business requirements and constraints., and translates them into executable system architecture. AI is applied as controlled logic across four layers.

2030			2035		
Technology		KPI	Technology	KPI	
Service Orchestration	• Calendar-driven Demand Orchestration Engine	CVR ≥23%	• Behavioral AI Orchestration	CVR ≥35%	
	• Service Reservation	Wait ≤6min	• Agent-Demand Control	Wait ≤3min	
	• Availability Matching Engine	Cancel ≤12%	• 4D Digital Twin	Cancel ≤5%	
	• Reservation-triggered Dispatch	Match ≤500ms [A]	• Travel-time Monetization	Matches≤100ms [A]	
	• Linkage API	Avail ≤500ms	• City Service Optimization	Avail ≤100ms	
Operation Control	• Customer Touchpoint PaaS	Res ≤1s	• Fleet Camera social sensing	Behavior ±1s	
	• Context-aware Notification	Cal ±5min	• IFTTT Trigger Dispatch	History ≤10s	
		Trigger ≤1s	• Autonomous Intent Agent	Reflect ≥98%	
	• Dispatch Optimization (VRP)	Trips ≥73/d [B]	• Integrated TMS	Trips ≥140/d [B]	
	• Real-time Routing	Load ≥14/d	• Fleet-cooperative Control	Load ≥112/d	
Fleet Vehicle	• Closed-loop Dispatch	Resp ≤8min	• Energy-Transport Optimizer	Resp ≤5min	
	• Demand Prediction	ETA err ≤3min	• Quantum-inspired Optimization	ETA err ≤2min	
	• Energy-aware Routing	Route ≤5s	• In / Out Energy Sync	Route ≤1s	
	• Demand-Supply Balancing	Alloc ≤10mi	• Real-time Replanning	Alloc ≤10s	
	• ETA Prediction	Fleet ±50m	• Constraint-aware Optimization	Fleet ±5m	
Distributed Infrastructure	• Capacity Allocation	Traffic ≤30s	• Multi-agent Coordination	Traffic ±1s	
	• ETA Feedback Control	Empty ≤25%	• Global State Synchronization	Empty ≤15%	
	• Queue Optimization	Util ≥75%	• Predictive Dispatch	Util ≥85%	
	• Swappable Vehicle Module	Util ≥87.5% [C]	• Dynamic Vehicle Coupling	Util ≥95.8% [C]	
	• MS-SOFC + SSB	Err ≤30cm	• Onboard Mill+ AI Air Bat	Err ≤5cm	
	• LH2 Tank TypeIV Subframe	MPC ≤50ms	• HTS coil winded LH2 Tank	MPC ≤10ms	
	• MgH2 MH Main chassis	Traj ≤50ms	• H- Conductor adapted MH	Traj ≤10ms	
	• PMSM Distributed Drive	Conv ≤100ms	• Real-time energy optimization	Conv ≤10ms	
	• SiC Inverter	Stability ±15%	• Thermal-compute coupling	Stability ±5%	
	• SbW (Dual Motor Redundant)	Fail-safe ≤100ms	• High-frequency power control	Coop ≥98%	
	• BbW (Electro-Hydraulic +Regen)	Percep ≥95%	• Quantum-enhanced LiDAR	Fail-safe ≤10ms	
	• CFRP Leaf + Active Suspension	Plan ≤200ms	• Quantum magnetometer	Perce ≥99.9%	
	• CFRP Spoke + CTIS Tire	Intv ≤1/100km	• Atonic clock	Plan ≤50ms	
				Intv ≤1/10,000km	
	• High-precision docking	Eff ≥75%	• Precision docking	Eff ≥90%	
	• Wind-driven Mill	Util ≥78%	• Ultra-low latency V2X sync	Util ≥96%	
	• Wind-driven AMR	Energy ±5% [D]	• Water-pipeline Mill	Energy ±1% [D]	
	• SOFC + Ammonia Tank	Compute ≤1s	• Pipeline-integrated AMR	Compute ≤10ms	
	• Superconducting Power Cable	Sync ≤1min	• PCFC	Sync ≤1s	
	• 6G Communication Network	Refuel ≤5min	• H-conductive BIMH Storage	Refuel ≤1min	
	• Standardized energy interface	S-D ≥0.78	• HTS-coiled LH2 Pipeline	S-D ≥0.96	
	• Low-latency communication	Load err ≤10%	• Quantum Communication	Load err ≤3%	
	• Sub-ms time synchronization	Var ≤±5%	• Energy-transport interface	Var ≤±1%	

I design systems where business requirements and technical execution are structurally aligned.

System Design for Mobility Feasibility

This page defines scenario design targets for Energy Efficiency = 0.58 (=58%), Utilization = 0.75 (=18h/day ÷ 24h/day), and Load Factor = 0.70 (=2.8pax ÷ 4pax). They are not proven outcomes; each target requires metered validation through integrated system architecture.

The design is structured across energy supply [kWh/day], vehicle dynamics and response [ms, cm, Hz], and control and computation [ms, Hz]. Net traction demand is 2,080.5kWh/day (=28.5kWh/trip × 73trips/day). At 60% electrical efficiency, required fuel-LHV input is 3,467.5kWh/day and the hydrogen-equivalent requirement is 104.1kg/day (=3,467.5 ÷ 33.3). Annual PV energy can exceed annual net demand, but an annual ratio does not prove 24-hour supply without power, storage, loss, and weather modelling.

Vehicle and control design ensure "motion sickness free" operation, defined by control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz. These are maintained through superconductivity enabled by liquid hydrogen cooling and real-time control via HPC operating at 100Hz (=1 ÷ 0.01sec). The HPC value is a design target and does not by itself prove continuous satisfaction or ODD expansion.

Block	Variable	Corrected value	Formula / definition	KPI link
KPI	Energy Efficiency	0.58 target	$\Sigma \text{ useful traction output} \div \Sigma \text{ purchased energy input}$	Energy
	Utilization	0.75 (=18÷24)	18h/day ÷ 24h/day	Operation
	Load Factor	0.70 (=2.8÷4)	2.8pax ÷ 4pax	Demand
Constraint	Operation Hours	24h/day	operating-window target	Utilization
	Stability	0.95	1 – 0.05	Energy
	Throughput	40veh/h	60min ÷ 1.5min	Infra
Energy	Net traction demand	2,080.5kWh/day	28.5×73	Energy
	Fuel-LHV input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC electrical efficiency	0.60	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	source constant	Energy
	H2-equivalent requirement	104.1kg/day	3,467.5÷33.3	Supply
	Annual net demand	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	scenario benchmark	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual energy only
	24h supply gate	Not proven by annual ratio	hourly power + storage + loss + weather model	Infra
	Supply loss	0.05 target	5%	Energy
	Supply stability	0.95 target	supplied ÷ required energy	Energy
Node	Node Delay	≤50ms	measured end-to-end node delay	Infra
	Dispatch Interval	5sec	60sec ÷ 12	Utilization
	Trips	73/day target	scenario target; capacity ceiling 40×12=480/day	Demand
	Fleet	13 vehicles	ceil(73÷6)	Operation
	Load capacity	4pax	2×2 seats	Demand
	Energy per trip	28.5kWh/trip	2,080.5÷73	Energy
	Continuous ratio	1.00 target	24÷24	Utilization
	Availability	0.99 target	available time ÷ scheduled time	KPI
	Reliability	0.99 target	successful cycles ÷ total cycles	Stability
Vehicle	Control Sync	≤20ms	100ms ÷ 5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m ÷ 10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec ÷ 10	Stability
	Response Delay	≤20ms	100ms ÷ 5	Motion sickness free
	Accel Jerk	≤0.5m/s²/s	1 ÷ 2	Motion sickness free
	Vibration	≤0.3Hz	1Hz ÷ 3.33	Motion sickness free
Control	Sync Accuracy	0.99	1 – 0.01	Stability
	Prediction Rate	1.00 target	successful predictions ÷ attempts	Operation
	Planning Rate	1.00 target	successful plans ÷ attempts	Operation
	Execution Rate	100Hz	1 ÷ 0.01sec	Control
	HPC Latency	≤10ms	measured inference latency	Stability
	Node Sync	≤50ms	measured synchronization delay	Stability
	Flow Capacity	40veh/h	60÷1.5	Infra
UX	Utilization Gain	0.10 target	0.75–0.65	KPI
Comparator	PEMFC efficiency	~60% direct H2	DOE technology comparison; not 22%	No exclusion on efficiency alone
	Battery downtime	4h/day assumption	24–20	Scenario only
	Battery utilization	0.83 assumption	20÷24	Scenario only
	ICE efficiency	0.30 assumption	scenario benchmark	Scenario only
Physics	Annual energy ratio	1.85	PV annual generation ÷ annual net demand	Energy
	Continuous supply	Validation pending	hourly balance required	Utilization
	Conversion loss	0.40	1–0.60	Energy
System	Feasibility gate	Not proven by multiplication	all independent safety, energy, operation, and financial gates must pass	Final
Conclusion	Hydrogen equivalent	104.1kg/day	2,080.5÷0.60÷33.3	Core
	Node operation	Validation pending	24h hourly model required	Core
	Utilization	0.75 target	18÷24	Core
	KPI status	Validation pending	not 0.58×0.75×0.70	Final

Feasibility Conditions for Autonomous Mobility System

This page does not present financial outcomes. It defines scenario conditions that must be independently validated. Energy Efficiency = 0.58, Utilization = 0.75, and Load Factor = 0.70 are design targets, not derived facts. Feasibility depends on energy supply [kWh/day], available power [kW], storage duration [h], vehicle response [ms], control accuracy [cm], ODD evidence, and commercial constraints.

Powertrain alternatives must be compared on duty-cycle efficiency, degradation, refuelling/charging downtime, mass, packaging, infrastructure, and total cost. Direct-hydrogen PEM fuel cells are not excluded by a 22% efficiency assumption; DOE lists approximately 60% electrical efficiency for direct-H2 PEM. MS-SOFC remains the selected base-case candidate for this concept, subject to full TCO, durability, start-up, emissions, and ODD validation.

Node operation requires 2,080.5kWh/day of net traction energy. At 60% efficiency the fuel-LHV requirement is 3,467.5kWh/day, equivalent to 104.1kg-H2/day. PV generation of 1,402,500kWh/year gives an annual energy ratio of approximately 1.85, but continuous operation still requires an hourly supply-demand, storage, conversion-loss, and weather model.

"Motion sickness free" refers to a condition where control latency $\leq 20\text{ms}$, decision latency $\leq 100\text{ms}$, positioning error $\leq 10\text{cm}$, and vibration $\leq 0.3\text{Hz}$ are simultaneously satisfied. This condition determines user acceptance and directly enables Utilization = 0.75 ($=18\text{h/day} \div 24\text{h/day}$). HPC maintains control at an execution rate of 100Hz ($=1 \div 0.01\text{sec}$), ensuring continuous satisfaction of response constraints. The definition is retained as the design requirement; compliance must be demonstrated by test evidence.

Block	Variable	Value / status	Formula / validation	KPI link
KPI	Energy Efficiency	0.58 target	metered useful output $\div$ purchased input	Energy
	Utilization	0.75 target	$18 \div 24$	Operation
	Load Factor	0.70 target	$2.8 \div 4$	Demand
Constraint	Operation Hours	24h/day target	operating window	Utilization
	Stability	0.95 target	$1 - 0.05$	Energy
	Throughput	40veh/h	1 vehicle $\div$ 1.5min	Infra
Node	Net energy	2,080.5kWh/day	28.5×73	Energy
	Fuel input	3,467.5kWh/day	$2,080.5 \div 0.60$	Energy
	SOFC efficiency	0.60 target	net electricity $\div$ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	constant	Energy
	H2 required	104.1kg/day	$3,467.5 \div 33.3$	Supply
	Annual net energy	759,383kWh/yr	$2,080.5 \times 365$	Energy
	PV capacity	1,500kW	$1.5\text{MW} \times 1,000$	Energy
	PV yield	1,100kWh/kW/yr	benchmark assumption	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	$1,500 \times 1,100 \times 0.85$	Energy
	Annual PV ratio	1.85	$1,402,500 \div 759,382.5$	Annual only
	Continuous supply	Not yet demonstrated	hourly balance + storage + reserve	Utilization
	Supply stability	0.95 target	supplied $\div$ required	Energy
	Node delay	$\leq 50\text{ms}$ target	measured	Infra
	Dispatch interval	5sec	$60 \div 12$	Utilization
	Trips	73/day target	scenario target	Demand
	Fleet	13	$\text{ceil}(73 \div 6)$	Operation
	Load	4pax	2×2	Demand
	Energy/trip	28.5kWh	$2,080.5 \div 73$	Energy
	Reliability	0.99 target	successful cycles $\div$ total	Stability
	Availability	0.99 target	available $\div$ scheduled time	KPI
Vehicle	Control Sync	$\leq 20\text{ms}$	$100\text{ms} \div 5$	Motion sickness free
	Decision Sync	$\leq 100\text{ms}$	100ms	Motion sickness free
	Position Error	$\leq 10\text{cm}$	$1\text{m} \div 10$	Motion sickness free
	Platoon Stability	$\leq 5\%$	0.05	Motion sickness free
	Control Cycle	100ms	$1\text{sec} \div 10$	Stability
	Response	$\leq 20\text{ms}$	$100\text{ms} \div 5$	Motion sickness free
	Jerk	$\leq 0.5\text{m/s}^2/\text{s}$	$1 \div 2$	Motion sickness free
	Vibration	$\leq 0.3\text{Hz}$	$1\text{Hz} \div 3.33$	Motion sickness free
	Sync Accuracy	0.99	$1 - 0.01$	Stability
Control	Prediction	1.00 target	successful prediction ratio	Operation
	Planning	1.00 target	successful plan ratio	Operation
	Execution	100Hz	$1 \div 0.01\text{sec}$	Control
	HPC latency	$\leq 10\text{ms}$ target	measured	Stability
	Node sync	$\leq 50\text{ms}$ target	measured	Stability
	Flow	40veh/h	$60 \div 1.5$	Infra
UX	Utilization Gain	0.10 target	$0.75 - 0.65$	KPI

This defines the conditions under which autonomous mobility may become operationally feasible. Passing one ratio does not substitute for validation of the remaining gates.

Proposed Responsibility Structure for Autonomous Mobility System

This is a proposed RACI hypothesis for discussion, not a statement of current accountability of any specific company or consortium. Final responsibility must be confirmed by corporate governance, contracts, permits, product-liability allocation, operational rules, and the approved safety case.

Continuous operation depends on explicit ownership across orchestration, fleet operation, vehicle/system safety, node execution, energy supply, commercial settlement, and finance. A method such as model-based development is not an accountable legal entity; each task must be assigned to an organization or contracted operator.

Domain	Function / task	Accountable (A)	Responsible (R)	Support / consulted	Status
Demand	Generate demand	Mobility Platform	Mobility Platform	AD Platform / partners	Proposed
	Capture demand	Mobility Platform	Mobility Platform	MaaS Operator	Proposed
	Set pricing	Commercial owner	Commercial owner	Finance / regulator	Contract required
	Approve pricing	Vehicle OEM / business owner	Commercial owner	Legal / regulator	Approval required
	Optimize load	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Manage UX	Mobility Platform	Mobility Platform	AD Platform / users	Proposed
	Define service	Vehicle OEM / business owner	Mobility Platform	MaaS Operator / user	Proposed
Operation	Execute dispatch	MaaS Operator	MaaS Operator	Mobility / AD Platforms	Permit + SLA
	Operate fleet	MaaS Operator	MaaS Operator	OEM / AD Platform	Permit + SLA
	Maintain service	MaaS Operator	MaaS Operator	Node operator / Mobility Platform	SLA
	Manage utilization	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Define hours	MaaS Operator	MaaS Operator	Regulator / node operator	Permit
	Control dispatch	MaaS Operator	AD control platform	Mobility Platform	Authority boundary
Vehicle	Define architecture	Vehicle OEM	Vehicle OEM	AD Platform / Tier-1	Product governance
	Supply platform	Vehicle OEM	Vehicle OEM	Tier-1 suppliers	Commercial contract
	Supply drivetrain	Vehicle OEM	Tier-1 supplier(s)	AD Platform	Commercial contract
	Package cabin	Vehicle OEM	Body integrator / supplier	Mobility Platform	Commercial contract
	Maintain vehicle	Fleet owner / MaaS Operator	Certified service provider	Vehicle OEM	SLA
	Realize control	Vehicle OEM	AD Platform + Tier-1	MaaS Operator	Safety case
	Maintain position	AD Platform / OEM	AD Platform	Mobility Platform	Safety case
Facility	Provide site	Node / Parking Operator	Node / Parking Operator	Vehicle OEM	Contract
	Operate facility	Node / Parking Operator	Node / Parking Operator	MaaS Operator	Permit + SLA
	Sell services	Commercial owner	Mobility Platform / provider	Node operator	Contract
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
	Manage circulation	Node operator	Node operator	Vehicle OEM	SLA
Service	Provide booking	Mobility Platform	Mobility Platform	AD Platform	Service governance
	Integrate service	Mobility Platform	Mobility Platform	Service providers	API contract
	Sell package	Commercial owner	Commercial owner	Finance	Contract
	Optimize stays	MaaS Operator	Mobility Platform	Service providers	Proposed
	Operate service	Service provider	Service provider	MaaS Operator	SLA
MaaS	Operate platform	MaaS Operator	AD + Mobility Platforms	Vehicle OEM	Authority boundary
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
External	Provide entry	External platform owner	External platform owner	Mobility Platform	API contract
	Send traffic	External platform owner	External platform owner	MaaS Operator	Commercial contract
	Receive traffic	MaaS Operator	Mobility Platform	External platform	Commercial contract
Energy	Define supply	Energy business owner	Energy integrator / supplier	AD Platform / node operator	Contract
	Produce hydrogen	Energy supplier	Energy supplier	Node operator	Permit + quality
	Distribute energy	Energy supplier	Energy integrator / supplier	Node operator	Contract
	Manage logistics	Energy Integrator	Energy Integrator	Energy supplier	Contract
	Operate energy	Node operator	Node operator	AD Platform	SLA
Software	Define scope	Vehicle OEM / product owner	AD Platform	Mobility Platform	Product governance
	Develop control	AD Platform	AD Platform	Vehicle OEM / Tier-1	Release governance
	Operate compute	AD Platform	Cloud / edge operator	MaaS Operator	SLA
	Maintain dispatch	MaaS Operator	AD + Mobility Platforms	Node operator	SLA
Finance	Collect mobility	Commercial owner	Finance / payment operator	Mobility Platform	Settlement contract
	Collect facility	Node operator	Finance / payment operator	Mobility Platform	Settlement contract
	Collect energy	Energy business owner	Finance / payment operator	Energy Integrator	Settlement contract
	Maintain projection	Business owner	Finance	Vehicle OEM	Governance
	Collect data	Data controller	Mobility Platform	AD Platform / operators	Privacy + data contract
	Collect facility data	Node operator	Mobility Platform	AD Platform	Privacy + data contract

Responsibility is complete only after the accountable legal entity, operational authority, safety authority, data controller, and revenue/cost owner are all confirmed.

Pricing Structure and Billing Rules for Autonomous Mobility System

The amounts below remain illustrative scenario tariffs, not verified market prices. To prevent double counting, one base trip fare is selected per trip; add-ons apply only when the corresponding service is consumed; subscriptions offset eligible base charges; B2B platform fees are charged only when not already embedded in another contract; dividends/revenue shares are settlements below revenue, not customer prices.

Category	Payer	Payee	Charge	Billing rule	Illustrative formula / annualization	P&L mapping
User	User	Service provider	Facility Fee	Add-on	¥500/visit×200/day×365 = ¥36.500M/yr	MaaS service
			Premium Service	Add-on	¥1,500/booking×120/day×365 = ¥65.700M/yr	Service revenue
			Add-on	Add-on	¥600/use×200/day×365 = ¥43.800M/yr	Service revenue
	MaaS Operator	Service Fee	Base fare (exclusive)	¥1,000/trip×130/day×365 = ¥47.450M/yr	Trip revenue	
		Trip Fee	Base fare (exclusive)	¥2,000/trip×100/day×365 = ¥73.000M/yr	Trip revenue	
		Premium Trip	Base fare (exclusive)	¥2,500/trip×90/day×365 = ¥82.125M/yr	Trip revenue	
		Subscription	Offsets eligible base fares	¥15,000/user×1,200 users = ¥18.000M/yr	Subscription revenue	
		Night Trip	Base fare (exclusive)	¥2,700/trip×80/day×365 = ¥78.840M/yr	Trip revenue	
		Parking Operator	Premium Facility	Add-on	¥800/visit×120/day×365 = ¥35.040M/yr	Facility revenue
			Loyalty Fee	Add-on / retention	¥200/visit×185/day×365 = ¥13.505M/yr	Service revenue
Facility AD Platform	Parking Operator	Site Rent	B2B fixed fee	¥300,000/month×30×12 = ¥108.000M/yr	Node OPEX	
		Operating Access	B2B access fee	¥700/visit×90/day×365 = ¥22.995M/yr	Node OPEX	
	Parking Operator	Service Fee	B2B service	¥300/visit×200/day×365 = ¥21.900M/yr	Node revenue	
	Service Provider	Parking Operator	Upgrade Rent	B2B fixed fee	¥2,000,000/month×30×12 = ¥720.000M/yr	Node OPEX
Service Service Provider	Mobility Platform	Booking Fee	Per booking	¥100/booking×150/day×365 = ¥5.475M/yr	Reservation fee	
		CRM Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee	
		API Fee	B2B usage fee	¥80/call×250/day×365 = ¥7.300M/yr	Platform fee	
		Premium UX	B2B product fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee	
	AD Platform	Optimization Fee	B2B product fee	¥50/call×300 calls/day×365 = ¥5.475M/yr	Platform fee	
		Premium API	B2B usage fee	¥150/call×60/day×365 = ¥3.285M/yr	Platform fee	
		Sync Fee	B2B product fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee	
	Service Provider	Late Fee	Customer charge	¥1,200/use×70/day×365 = ¥30.660M/yr	Service revenue	
	Mobility MaaS Operator	AD Platform	Control Fee	B2B product fee	¥100/dispatch×100/day×365 = ¥3.650M/yr	Control OS
Routing Fee			B2B product fee	¥120/dispatch×110/day×365 = ¥4.818M/yr	Platform fee	
Mobility Platform		UX Fee	B2B product fee	¥50/trip×100/day×365 = ¥1.825M/yr	Platform fee	
		Parking Operator	Advanced Control	¥150/dispatch×90/day×365 = ¥4.928M/yr	Node service	
AD Platform		MaaS Operator	Incentive	Contra-revenue / settlement	¥200,000/month×30×12 = ¥72.000M/yr	Settlement; not gross revenue
User		MaaS Operator	Surge Pricing	Base fare adjustment	¥2,800/trip×80/day×365 = ¥81.760M/yr	Trip revenue
		Off-peak Discount	Base fare adjustment	¥1,600/trip×70/day×365 = ¥40.880M/yr	Trip revenue	
Platform MaaS Operator	AD Platform	OS Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee	
		Monthly OS	B2B fixed fee	¥250,000/month×30×12 = ¥90.000M/yr	Platform fee	
	Service Provider	AD Platform	Node Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
		Data Fee	B2B usage fee	¥100,000/month×30×12 = ¥36.000M/yr	Platform fee	
	Parking Operator	AD Platform	Facility Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
		Analytics Fee	B2B fixed fee	¥120,000/month×30×12 = ¥43.200M/yr	Platform fee	
		Control OS	B2B product fee	¥220/trip×110/day×365 = ¥8.833M/yr	Control OS	
		Booking Link	B2B usage fee	¥180,000/month×30×12 = ¥64.800M/yr	Reservation fee	
External External Platform	AD Platform	API Fee	B2B usage fee	¥120/use×5,000/day×365 = ¥219.000M/yr	Platform fee	
		IoT Fee	B2B usage fee	¥90/booking×200/day×365 = ¥6.570M/yr	Platform fee	
		Tx Fee	B2B usage fee	¥200/booking×100/day×365 = ¥7.300M/yr	Platform fee	
		Premium API	B2B usage fee	¥150/call×6,000/day×365 = ¥328.500M/yr	Platform fee	
		Premium Tx	B2B usage fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee	
		Premium Referral	Revenue share / settlement	15% of eligible booking revenue	Settlement; not gross revenue	
		Sync Fee	B2B usage fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee	
Energy AD Platform	Finance Co.	Energy Fee	B2B service	20kWh/trip×100/day×¥30×365 = ¥21.900M/yr	Energy revenue	
		Peak Energy	B2B service	25kWh/trip×90/day×¥45×365 = ¥36.956M/yr	Energy revenue	
		Smart Energy	B2B service	¥70/use×120/day×365 = ¥3.066M/yr	Energy revenue	
	Service Provider	AD Platform	Energy Opt	B2B product fee	¥50/use×150/day×365 = ¥2.738M/yr	Platform fee
Finance MaaS Operator	Finance Co.	Base Energy	B2B settlement	18kWh/trip×110/day×¥28×365 = ¥20.236M/yr	Energy OPEX	
		Lease	B2B lease	¥3,000,000/year×365-day basis = contract specific	Lease / CAPEX	
		Dividend	Settlement only	profit distribution after tax	Not revenue	
		Node Lease	B2B lease	¥4,500,000/year per node	Lease / CAPEX	
		Node Lease	B2B lease	¥2,500,000/year per node	Lease / CAPEX	
	Service Provider	Finance Co.	Lease	B2B lease	¥2,000,000/year×365-day basis = contract specific	Lease / CAPEX
		Dividend	Settlement only	profit distribution after tax	Not revenue	
		Service Lease	B2B lease	¥2,500,000/year per asset	Lease / CAPEX	

Only lines explicitly mapped to the P&L may be included. Settlement lines and mutually exclusive base fares must not be added again.
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Autonomous Mobility Business Control Structure

System conditions define the feasibility of control and set operational boundaries. Values below are phase targets or scenario assumptions; they are not externally verified outcomes. Minimum gates on P1 and phase targets here are different by design and must be labelled separately.

Revenue is calculated with unit-consistent formulas. When fare is per passenger, annual revenue = Trips/day × Seats × Load factor × Fare/passenger × 365. Break-even trips/day cannot be calculated without a defined annual controllable-cost allocation.

Layer	Control item	Decision logic	2030	2035	P&L impact
Demand	Demand generation	Requests ÷ potential demand	≥0.50	≥0.70	Trips
	Conversion	Bookings ÷ requests	≥0.70	≥0.85	Trips
	Matching	Matched ÷ requests	≥0.85	≥0.95	Trips
	Trip realization	Executed ÷ matched	≥0.80	≥0.95	Trips
	Stabilization	Matched trips + planned demand	≥0.80	≥0.95	Trips
	Time synchronization	Calendar-based demand	≥0.60	≥0.85	Trips
	Service capture	Reservation-based trips ÷ total	≥0.50	≥0.75	Trips
	Demand-cap control	Demand ≤ system capacity	100%	100%	Trips cap
Operation	Trips	Executed trips/day	73	140	Trips
	Utilization	Operating time ÷ 24h	≥0.75	≥0.85	Trips
	Dispatch success	Successful dispatch ÷ requests	≥0.95	≥0.99	Trips
	Supply-body cycle	Active body ÷ total cycle	≥0.80	≥0.95	Utilization
	Waiting time	Passenger request-to-pickup	≤360sec	≤180sec	Trips
	Turnaround	Idle + service time	≤25min	≤10min	Utilization
	Queue control	Total wait ÷ requests	≤300sec	≤120sec	Conversion
	Optimization	Served ÷ queued	≥0.90	≥0.98	Trips
	Scheduled continuity	Continuous-operation ratio	≥0.80	≥0.95	Utilization
	Operation cap	Operation ≤24h	Fixed	Fixed	Cap
Vehicle	Latency control	Control delay	≤100ms	≤50ms	Dispatch
	Stability	Deviation ÷ target	≤15%	≤5%	Safety
	Intervention	Interventions ÷ km	≤0.010/km	≤0.0001/km	Utilization
	Motion response	Control latency / planning / ride targets	Design gate	Design gate	Utilization
	Response accuracy	Correct ÷ total control	≥0.90	≥0.98	Dispatch
	Platoon composition	Vehicles per formation	≥1.5	≥3.0	Load
	Ride continuity	Continuous ride + trips	≥0.85	≥0.90	Utilization
	Thermal constraint	Temperature limit	Fixed	Improved	Operation
	Vehicle availability	Available fleet ÷ total	≥0.80	≥0.95	Trips
Energy	End-to-end efficiency	Useful output ÷ purchased input	0.58 target	0.65 target	Cost
	Energy cost/trip	Cost ÷ trips	≤¥500	≤¥350	Cost
	Supply stability	Continuous supplied ÷ required	≥0.95	≥0.99	Utilization
	H2-equivalent requirement	Net energy ÷ efficiency ÷33.3	104.1kg/day	171.2kg/day	Trips cap
	PV role	Annual energy contribution	Annual ratio only	Annual ratio only	Cost
	Storage capacity	Hourly storage / reserve	Required	Required	Trips cap
	Distribution	Loss + supply	≤10%	≤5%	Cost
	Node allocation	Allocated ÷ demand	≥0.90	≥0.98	Trips
	Energy scheduling	Scheduled + required	≥0.85	≥0.95	Trips
System	Trip generation	Demand×Matching×Dispatch×Utilization	73/day	140/day	Revenue
	Load optimization	Passengers ÷ capacity	0.70	0.80	Revenue
	Fare	JPY per passenger	¥800	¥900	Revenue
	Throughput	Trips ÷ hour	≥1.5	≥3.0	Revenue
	Density	Trips ÷ area	Low	High	Revenue
	Scaling factor	Trips 2035 ÷ 2030	1.00×	1.92×	Revenue
	Energy linkage	28.5kWh × trips	2,080.5kWh	3,990kWh	Cost
	Hydrogen linkage	Energy ÷ efficiency ÷33.3	104.1kg	171.2kg	Cost
	Fleet sizing	ceil(trips ÷6)	13	24	Trips cap
	Congestion	Load ÷ capacity	Medium	High	Limit
	Node capacity	Throughput ≤ capacity	Limited	Expanded	Cap
Financial	Annual passenger revenue	Trips×Seats×Load×Fare×365	¥59.685M	¥183.960M	Profit
	Revenue per vehicle trip	Seats×Load×Fare	¥2,240	¥3,600	Revenue
	Contribution	Allocated revenue – allocated cost	See rates	See rates	IRR
	Break-even trips/day	Annual controllable cost ÷ (Seats×Load×Fare×365)	TBD	TBD	Profit
	Cost control	Energy + operation + vehicle + infrastructure	High	Reduced target	Cost
	Price constraint	Applicable fare / contract rules	Validate	Validate	Revenue
	IRR hurdle	Return target	≥10%	≥12%	Investment

This defines unit-consistent control logic connecting operational conditions to trips, revenue, cost, and investment gates.

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Autonomous Mobility Business Profit & Loss Structure (2030 / 2035)

This is a scenario P&L, not a forecast. Each monetary line is recalculated from the displayed formula. Non-monetary operating drivers are separated from revenue. The prior ¥36B node-sale line is excluded from the base case because asset-transfer price, delivery cost, and COGS were not defined; recurring node-operation service remains included.

Layer	Category	Item	2030 formula	2030	2035 formula	2035
Revenue	Energy	Elec Internal	$4.0M \times 20\% \times 50\% \times 1,200 \times \yen 15$	¥7.200B	$4.0M \times 30\% \times 50\% \times 1,500 \times \yen 15$	¥13.500B
		Elec External	$4.0M \times 25\% \times 50\% \times 1,200 \times \yen 15$	¥9.000B	$4.0M \times 35\% \times 50\% \times 1,500 \times \yen 15$	¥15.750B
		H2 Internal	$4.0M \times 80\% \times 1\% \times 67 \times \yen 333$	¥0.714B	$4.0M \times 70\% \times 20\% \times 67 \times \yen 333$	¥12.494B
		H2 External	$4.0M \times 75\% \times 1\% \times 67 \times \yen 333$	¥0.669B	$4.0M \times 65\% \times 20\% \times 67 \times \yen 333$	¥11.602B
		Titanium	$80,000 \times 0.3 \times \yen 72,000$	¥1.728B	same	¥1.728B
		HPC Compute Service	$600 \text{ GPU} \times 8,760 \times 35\% \times \yen 700$	¥1.288B	$5,400 \text{ GPU} \times 8,760 \times 55\% \times \yen 600$	¥15.610B
Driver		24H Utilization	$18 \div 24$	0.75	$20.4 \div 24$	0.85
		Load Factor	$2.8 \div 4$	0.70	$4.0 \div 5$	0.80
		Swap-body Util.	Active÷Total	0.80	Dynamic composition	0.95
		Night Operation	Night revenue÷Total	0.35	Continuous operation	0.60
		Auto Loading Rev.	Loading fee rev.÷Total	0.22	Integrated logistics	0.35
Revenue	MaaS	Night Passenger	$25 \times 24 \times \yen 15,000 \times 365 \times 60\%$	¥1.971B	$500 \times 24 \times \yen 15,000 \times 365 \times 60\%$	¥39.420B
		Night Cargo	$25 \times 24 \times \yen 5,000 \times 365 \times 40\%$	¥0.438B	$500 \times 24 \times \yen 5,000 \times 365 \times 40\%$	¥8.760B
		Mobile Storage	$400 \times \yen 40,000 \times 12$	¥0.192B	$5,000 \times \yen 40,000 \times 12$	¥2.400B
	Platform	Auto Loading	$269,000 \times \yen 22,300$	¥5.999B	$538,000 \times \yen 22,300$	¥11.997B
		Platform Fee	Dispatch/API fee	¥0.800B	Integrated OS fee	¥6.500B
		Reservation Fee	Booking linkage	¥0.350B	Integrated booking	¥2.200B
		Node Service	Node operation fee	¥0.420B	Node ecosystem service	¥4.800B
Total		Total Revenue	sum monetary revenue	¥30.769B	sum monetary revenue	¥146.761B
OPEX	Energy	H2 Internal COGS	$4.0M \times 80\% \times 1\% \times 67 \times \yen 150$	▲¥0.322B	$4.0M \times 70\% \times 20\% \times 67 \times \yen 150$	▲¥5.628B
		H2 External COGS	$4.0M \times 75\% \times 1\% \times 67 \times \yen 150$	▲¥0.302B	$4.0M \times 65\% \times 20\% \times 67 \times \yen 150$	▲¥5.226B
		Elec Internal	$4.0M \times 20\% \times 50\% \times 1,200 \times \yen 6.7$	▲¥3.216B	$4.0M \times 30\% \times 50\% \times 1,500 \times \yen 6.7$	▲¥6.030B
		Elec External	$4.0M \times 25\% \times 50\% \times 1,200 \times \yen 6.7$	▲¥4.020B	$4.0M \times 35\% \times 50\% \times 1,500 \times \yen 6.7$	▲¥7.035B
		Titanium	$80,000 \times 0.3 \times \yen 6,000$	▲¥0.144B	same	▲¥0.144B
		Material Cycle	scenario cost	▲¥0.300B	onboard circulation	▲¥1.800B
		Node Optimization	distributed control	▲¥0.120B	integrated mesh	▲¥0.950B
		Energy Sync	In/Out sync	▲¥0.080B	real-time optimization	▲¥0.600B
	MaaS	MaaS Ops	$25 \times \yen 81.0M$	▲¥2.025B	$500 \times \yen 45.6M$	▲¥22.800B
		Dispatch Ops	fleet dispatch	▲¥0.550B	integrated dispatch	▲¥4.200B
		Remote Ops	remote support	▲¥0.300B	integrated operation	▲¥2.100B
	Platform	Platform Ops	¥1.0B/year	▲¥1.000B	¥6.0B/year	▲¥6.000B
		Cloud Ops	base + power + service	▲¥0.731B	base + power + service	▲¥6.680B
		AI Ops	AI orchestration	▲¥0.220B	integrated AI control	▲¥2.400B
Total		Total OPEX	sum	▲¥13.329B	sum	▲¥71.593B
SG&A	Energy	Energy SG&A	energy product revenue×10%	▲¥1.931B	energy product revenue×10%	▲¥5.507B
		Maintenance	scenario	▲¥0.250B	integrated maintenance	▲¥1.800B
	MaaS	MaaS SG&A	MaaS revenue×15%	▲¥0.390B	MaaS revenue×15%	▲¥7.587B
		Customer Ops	user support	▲¥0.180B	integrated support	▲¥1.500B
	Platform	Platform SG&A	SW revenue×20%	▲¥1.514B	SW revenue×20%	▲¥5.099B
		Partner Ops	alliance operation	▲¥0.120B	integrated alliance	▲¥1.100B
Total		Total SG&A	sum	▲¥4.385B	sum	▲¥22.594B
Profit	Total	EBITDA	Revenue–OPEX–SG&A	¥13.054B	Revenue–OPEX–SG&A	¥52.574B
Dep.	Energy/MaaS/Platform	Node	$20.000B \div 15y$	▲¥1.333B	$180.000B \div 15y$	▲¥12.000B
		GTFC	$18.450B \div 15y$	▲¥1.230B	$33.450B \div 15y$	▲¥2.230B
		Mill	$0.600B \div 10y$	▲¥0.060B	$15.600B \div 10y$	▲¥1.560B
		Battery	$5.000B \div 10y$	▲¥0.500B	$10.000B \div 10y$	▲¥1.000B
		Magnetic	$3.000B \div 10y$	▲¥0.300B	$6.000B \div 10y$	▲¥0.600B
		HTS	$4.000B \div 10y$	▲¥0.400B	$8.000B \div 10y$	▲¥0.800B
		Water	$2.000B \div 10y$	▲¥0.200B	$4.000B \div 10y$	▲¥0.400B
		HPC	$5.000B \div 15y$	▲¥0.333B	$45.000B \div 15y$	▲¥3.000B
		Energy Device	Mill/Batt/HTS group	▲¥0.600B	Integrated energy device	▲¥3.600B
		Vehicle Dev	$50.000B \div 5y$	▲¥10.000B	$100.000B \div 5y$	▲¥20.000B
		Prototype	$10.000B \div 5y$	▲¥2.000B	$10.000B \div 5y$	▲¥2.000B
		Swap-body Dev	Swap-body architecture	▲¥1.500B	Dynamic coupling	▲¥5.200B
		Platform Dev	$40.000B \div 5y$	▲¥8.000B	$110.000B \div 5y$	▲¥22.000B
		AI Platform	AI orchestration	▲¥1.800B	Integrated AI infra	▲¥8.500B
		Control OS	Dispatch/control OS	▲¥1.200B	City OS integration	▲¥6.200B
		Infra Sync	Node synchronization	▲¥0.450B	Integrated node mesh	▲¥2.600B
		Revenue Logic	24H utilization logic	▲¥0.300B	Dynamic load logic	▲¥1.500B
Total		Total Depreciation	sum	▲¥30.206B	sum	▲¥93.190B
Profit	Total	Operating Profit	EBITDA–Dep.	▲¥17.152B	EBITDA–Dep.	▲¥40.616B
		Contribution before depreciation	Revenue–OPEX–SG&A	¥13.054B	Revenue–OPEX–SG&A	¥52.574B
		Operating Margin	Operating Profit÷Revenue	-55.7%	Operating Profit÷Revenue	-27.7%
		Project Payback	see cash-flow table	not achieved	see cash-flow table	discounted not achieved by 2035
		Project IRR	reconciled cash flow	-12.13%	12% hurdle	not met

Assumptions retained unless corrected above: city 1/5; MaaS vehicles 25/500; nodes 4/36; passenger cars 4.0M/year; trucks 80,000/year; electricity sales ¥15/kWh and cost ¥6.7/kWh; H2 sales ¥333/kg and cost ¥150/kg; titanium exchange 80,000×0.3, sales ¥72,000 and cost ¥6,000. These are scenario assumptions requiring primary-source validation.

2030 Vehicle Platform

A 12m autonomous MaaS skateboard uses an MS-SOFC scenario with city gas 70% and mill-derived hydrogen 30%. Hydrogen is supplied from stationary/node-side milling; batteries support transient power. Component ranges sum to approximately 9.8–15.1t before the stated 10–15% integration reserve; the upper case approaches 17.3t. Therefore, payload is not claimed until body, fluids, passengers, wiring/piping, axle loads, and certified curb mass are included. The 10–15kg LH2 buffer requires approximately 340L after the 1.6 packaging factor. The envelope is corrected to 450×550×1,400mm (346.5L external envelope). For 240kW electrical output at 60% efficiency, fuel input is 400kW and nominal rejected heat is approximately 160kW, not 96kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		12m flat skateboard	-	upper container unused	-	-	overall
Total energy demand	1	1.2kWh/km	$1.2 \times 1,000$	1,200kWh	-	-	overall
Required fuel energy	1	MS-SOFC 60%	$1,200 \div 0.60$	2,000kWh-LHV	-	-	overall
Hydrogen ratio	1	H2 30% blend	$2,000 \times 0.30$	600kWh-H2	-	-	overall
Required H2	1	33.3kWh/kg	$600 \div 33.3$	18.0kg-H2	-	-	overall
City-gas ratio	1	70%	$2,000 \times 0.70$	1,400kWh-gas	-	-	overall
LH2 buffer	1	10–15kg	reserve	10–15kg-H2	450/550/1400	150–250kg	left
LH2 installed volume	1	insulation×1.6	$15 \div 70.8 \times 1.6$	≈340L	450/550/1400	included	left
City-gas tank	2	blended-fuel support	-	operation reserve	300/400/1600	300–500kg	left
Water tank	2	cooling/process water	-	≈180L	250/400/1200	180kg	left
Process-water control	2	steam-to-carbon / condensate	validated setpoint	not NOx claim	180/250/400	80–120kg	front
O2 membrane	2	optional O2 management	test gate	efficiency TBD	180/250/450	80–120kg	front
CO2 membrane	2	reformate cleanup	70–90% target	CO2 removal target	220/250/500	100–150kg	rear
Steam membrane	2	H2O separation	test gate	reforming support	180/250/450	80–120kg	rear
Reformer	2	steam reforming	fuel-path model	H2+CO reformate	260/350/700	250–350kg	front
Gas/liquid separator	2	50L/min	50×2	100L/min	220/250/450	120–180kg	F/R
H2 purifier	2	50L/min	50×2	100L/min	220/250/500	120–180kg	F/R
Vaporizer	2	20kg/h	20×2	40kg/h	240/300/600	120–180kg	left
LH2 pump	2	50L/min	50×2	100L/min	180/220/350	60–100kg	left
Pressure regulator	4	high→low P	-	4 systems	120/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	120/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1400	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		MS-SOFC power control	-	battery transient	-	-	center
MS-SOFC modules	6	40kW/unit	40×6	240kW	320/1000/1400	1,200–1,800kg	center
MS-SOFC efficiency	1	electrical efficiency	-	≈60% target	-	-	center
Inverter	2	120kW	120×2	240kW	180/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	160/350/500	120–180kg	upper center
Solid-state battery	1	400kWh	-	400kWh	260/1200/2200	2,500–3,500kg	center
HV J-BOX	1	HV distribution	-	1 system	180/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	120/250/300	40–80kg	center
Radiator	2	rejected heat	$240 \times (1/0.60 - 1)$	≈160kW	180/500/1400	180–300kg	front
Cooling pump	2	50L/min	50×2	100L/min	160/180/250	40–80kg	overall
Waste-heat recovery	1	Rankine/thermal	validated recovery	aux. power TBD	260/600/900	250–450kg	rear
Insulation monitoring	1	HV safety	-	1 system	80/200/300	<20kg	center
Power protection	1	fuse/breaker	-	integrated	80/200/300	<20kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	prediction/planning	latency≤100ms	control target	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 Drivetrain		electric drive	-	AD ride stability	-	-	corners
IWM	4	80kW/wheel	80×4	320kW	320/500/500	1,000–1,400kg	corners
Regen brake	4	regenerative	-	4 wheels	400/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	400/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	260/320/420	250–350kg	corners
Air suspension+CFRP	4	vibration suppression	-	4 wheels	260/350/450	250–350kg	corners
CTIS tire	4	pressure control	-	4 wheels	220/500/500	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	550–600mm target	-	-	overall
Crossmember	6	structural beam	-	6 beams	160/1600/160	250–400kg	lateral
MH main chassis	2	H2 storage+frame	-	≈10kg-H2	400/350/5500	400–800kg	longitudinal
LH2 subframe	2	cooling+support	-	2 systems	260/450/2200	300–500kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	50–100/car	400–700kg	lower/center
Container lock	4	swap fixation	-	4 points	180/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	180/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	gas/LH2/water	150/500/900	<120kg	L/R
Boarding assist	1	kneel+2-step	–100mm	access assist	220/300/800	120–200kg	right
Payload margin	1	25t GVW–verified curb mass	full integration required	TBD; prior 10–13t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

2035 Vehicle Platform

A 12m platform targets a 380–400mm low floor. The platform-weight and 13–16t payload claims are removed pending full integrated mass, body, fluids, passengers, wiring/piping, axle-load, and certification data. The energy balance now separates the 400kWh air-battery contribution from PCFC energy demand.

PCFC-only hydrogen demand is 48.1kg/day (=1,200÷33.3÷0.75). With a 400kWh air-battery contribution, net PCFC energy is 800kWh and net H2 demand is 32.0kg/day. Six mills at 0.865kg/h for 12.5h produce 64.9kg/day, leaving a 32.9kg/day gross surplus before process losses. The 10–15kg LH2 buffer still requires about 340L; the envelope is corrected to 350×850×1,200mm (357L). Rejected heat at 240kW and 75% efficiency is approximately 80kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		system demand	1.2×1,000	1,200kWh	-	-	overall
Gross H2 before air battery	1	PCFC 75%	1,200÷(33.3×0.75)	48.1kg-H2	-	-	overall
Air-battery energy	4	100kWh/unit	100×4	400kWh	120/350/1200	≈849kg	right
H2 reduction equivalent	1	25kWh/kg-H2	400÷25	16.0kg-H2 eq.	-	-	right
Net PCFC energy	1	1,200–400	-	800kWh	-	-	overall
Net H2 required	1	PCFC 75%	800÷(33.3×0.75)	32.0kg-H2	-	-	overall
Hydrogen generation	6	0.865kg/h	0.865×6×12.5	64.9kg-H2	240/450/650	720–1,080kg	F/R
Gross surplus	1	before process losses	64.9–32.0	32.9kg-H2	-	-	overall
LH2 buffer	1	10–15kg reserve	-	10–15kg-H2	350/850/1200	100–150kg	left
LH2 installed volume	1	insulation×1.6	15÷70.8×1.6	≈340L	350/850/1200	included	left
Water tank	6	9kg/kg-H2 gross	64.9×9	584L	200/400/1200	584kg	left
Slurry tank	6	water×1.4	584×1.4	818L	230/450/1300	980–1,227kg	left
Titanium cartridge	6	3kg/0.865kg-H2	64.9÷0.865×3	≈225kg	160/300/800	225kg	L/R
PCFC modules	6	40kW/unit	40×6	240kW	180/900/1200	720–1,080kg	upper center
PCFC efficiency	1	electrical target	-	75% target	-	-	center
AMR liquefier	2	surplus-H2 process	10kg/day×2	process target	240/450/800	250–400kg	F/R
H2 purifier	3	50L/min	50×3	150L/min	180/250/500	120–180kg	F/R
Gas/liquid separator	3	50L/min	50×3	150L/min	200/250/450	120–180kg	F/R
Process-water management	2	condensate / humidity balance	validated setpoint	closed-loop target	150/250/400	80–120kg	front
O2 management membrane	2	optional cathode control	test gate	efficiency TBD	150/250/450	80–120kg	front
CO2 membrane	0	direct-H2 baseline	not required	excluded	-	0kg	rear
Steam separator	2	H2O separation	water balance	condensate recovery	150/250/450	80–120kg	rear
Reformer	0	direct-H2 baseline	not required	excluded	-	0kg	front
Anode loop	1	fuel recirculation	20% target	efficiency gain target	180/250/500	80–120kg	center
LH2 pump	2	50L/min	50×2	100L/min	150/220/350	60–100kg	left
Vaporizer	2	20kg/h	20×2	40kg/h	180/300/600	100–160kg	left
Pressure regulator	4	high→low P	-	4 systems	100/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	100/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1200	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		PCFC power control	-	battery transient	-	-	center
Inverter	2	120kW	120×2	240kW	150/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	130/350/500	120–180kg	upper center
Response battery	1	80kWh	-	80kWh	120/900/1600	350–500kg	upper center
HV J-BOX	1	HV distribution	-	1 system	150/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	100/250/300	40–80kg	center
Radiator	2	rejected heat	240×(1/0.75–1)	≈80kW	120/400/1200	120–200kg	front
Cooling pump	2	50L/min	50×2	100L/min	120/180/250	40–80kg	overall
HTS cooling line	1	LH2 cooling	-	1 system	60/60/12000	80–120kg	frame
HTC/HTS cable	1	low-loss power	-	1 system	50/50/12000	60–120kg	center→wheel
Waste-heat recovery	1	TE/TPV/Rankine	validated recovery	aux. power TBD	220/500/800	200–350kg	rear
Waste-heat utilization	1	efficiency gain	-	target only	-	-	rear
Power protection	1	insulation/fuse	-	integrated	80/200/300	<30kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	planning/control compute	validated ODD load	performance TBD by evidence	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 HTS Drivetrain		SC drive+ride ctrl	-	no-sick UX	-	-	corners
LH2 SC-IWM	4	80kW/wheel	80×4	320kW	260/450/450	800–1,200kg	corners
SC regen brake	4	regen braking	-	4 wheels	300/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	300/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	220/300/400	200–300kg	corners
Air sus.+CFRP	4	vibration cut	-	4 wheels	220/300/400	200–300kg	corners
CTIS/CFRP wheel	4	pressure+light wheel	-	4 wheels	200/450/450	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	380–400mm target	-	-	overall
HTS crossmember	6	beam+cooling	-	6 beams	120/1600/120	200–300kg	lateral
MH main chassis	2	H2 storage+frame	-	≈20kg-H2	300/300/5500	300–600kg	longitudinal
LH2/HTS subframe	2	cooling+support	-	2 systems	200/400/2000	200–400kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	30–80/vehicle	300–600kg	lower/center
Container lock	4	swap fixation	-	4 points	150/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	150/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	water/Ti/slurry	120/400/800	<100kg	L/R
Boarding assist	1	kneel+2-step	–100mm	access assist	200/300/800	120–200kg	right
Payload margin	1	25t GVW-verified curb mass	full integration required	TBD; prior 13–16t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

IRR and Cash Flow Structure

This page adds externally sold distributed-HPC capacity to the reconciled base cash flow. The node plan is 4/8/12/20/28/36; existing HPC CAPEX of ¥5B/¥45B implies ¥8.33M per installed unit and 150 H100-equivalent accelerators/node. Paid external utilization rises from 35% to 55% and blended price declines from ¥700 to ¥600 per accelerator-hour. Internal mobility compute is excluded from revenue. Incremental HPC OPEX includes 1.2kW per accelerator, PUE 1.15, ¥15/kWh electricity and service cost equal to 15% of HPC revenue. The case remains pre-tax and unlevered; working capital and cash taxes are zero. IRR is the discount rate that sets NPV to zero. Project payback is the first year in which cumulative nominal FCF becomes non-negative. Discounted payback is assessed separately. Node payback cannot be calculated because node-level revenue, OPEX, CAPEX, tax, and renewal cash flow are not separately specified.

Investment and Depreciation Structure			
Investment / Instrument	Timing	Dep.	Purpose
Initial Development Investment (2025–2030)			
Platform / AI / Urban OS	2025–2030	5y	Core control platform
MaaS Vehicle Prototype	2025–2028	5y	Vehicle feasibility
HTS / Battery / Mill	2026–2030	10y	Energy feasibility
Initial Nodes (4)	2027–2030	15y	Operational deployment
HPC / GTFC	2027–2030	15y	Control infrastructure
Expansion Investment (2030–2035)			
Node expansion (32; 36 total)	2030–2035	15y	City-scale deployment
Fleet Expansion (500)	2030–2035	5y	Utilization scaling
Distributed Energy Infrastructure	2030–2035	10–15y	Hydrogen/electricity scaling
AI / Dispatch Scaling	2030–2035	5y	Real-time optimization
Node Synchronization Infrastructure	2032–2035	15y	Distributed control

Displayed P&L Reference					
Year	Revenue	EBITDA	Operating Profit	EBITDA Margin	Operating Margin
2030	¥30.769B	¥13.054B	▲¥17.152B	42.4%	-55.7%
2035	¥146.761B	¥52.574B	▲¥40.616B	35.8%	-27.7%

Annual Cash Flow and Discounted Value								
Cash-flow formulas: FCF _t = EBITDA _t – CAPEX _t – ΔWC _t – Cash Taxes _t ; DCF _t = FCF _t / (1+r) ^t ; IRR solves Σ FCF _t / (1+IRR) ^t = 0.								
Year	Revenue	EBITDA	CAPEX	FCF	Discount factor 8%	Discounted FCF	Cumulative nominal FCF	Cumulative discounted FCF
2025	¥0.000B	▲¥5.000B	▲¥35.000B	▲¥40.000B	1.000	▲¥40.000B	▲¥40.000B	▲¥40.000B
2026	¥0.000B	▲¥6.000B	▲¥40.000B	▲¥46.000B	0.926	▲¥42.593B	▲¥86.000B	▲¥82.593B
2027	¥2.000B	▲¥5.000B	▲¥45.000B	▲¥50.000B	0.857	▲¥42.867B	▲¥136.000B	▲¥125.460B
2028	¥8.000B	▲¥2.000B	▲¥40.000B	▲¥42.000B	0.794	▲¥33.341B	▲¥178.000B	▲¥158.800B
2029	¥18.000B	¥3.000B	▲¥35.000B	▲¥32.000B	0.735	▲¥23.521B	▲¥210.000B	▲¥182.321B
2030	¥30.769B	¥13.054B	▲¥28.000B	▲¥14.946B	0.681	▲¥10.172B	▲¥224.946B	▲¥192.493B
2031	¥52.653B	¥19.893B	▲¥22.000B	▲¥2.107B	0.630	▲¥1.328B	▲¥227.053B	▲¥193.821B
2032	¥74.761B	¥26.910B	▲¥20.000B	¥6.910B	0.583	¥4.032B	▲¥220.143B	▲¥189.789B
2033	¥98.430B	¥35.199B	▲¥18.000B	¥17.199B	0.540	¥9.292B	▲¥202.944B	▲¥180.497B
2034	¥122.583B	¥43.880B	▲¥15.000B	¥28.880B	0.500	¥14.447B	▲¥174.064B	▲¥166.050B
2035	¥146.761B	¥52.574B	▲¥12.000B	¥40.574B	0.463	¥18.794B	▲¥133.489B	▲¥147.256B

Financial Results and Return Assessment			
Financial result	Value	Interpretation	Decision meaning
Total CAPEX 2025–2035	¥310.000B	Sum of annual CAPEX	Pre-tax / unlevered model
Project IRR	-12.13%	Below 12% hurdle	Return hurdle not met
NPV @ 8%	▲¥147.256B	Discounted value	8% return-hurdle gap
NPV @ 12%	▲¥147.160B	Hurdle-rate value gap	Value uplift required
Nominal project payback	2039 (reference)	2035 FCF held flat	No renewal CAPEX/tax/WC
Discounted payback	Not achieved	Cumulative DCF remains negative	Separate scenarios required
Node payback	Not calculated	Node-level cash flow absent	Add node economics
2035 EBITDA Margin	35.8%	Displayed P&L	Cash generation before D&A
2035 Operating Margin	-27.7%	Displayed P&L	Depreciation-heavy loss

Sensitivity to Cash Flow Improvement					
Sensitivity case	Explicit cash-flow change	IRR	NPV @ 8%	NPV @ 12%	Interpretation
No HPC external sales	HPC revenue excluded	-17.76%	▲¥165.852B	▲¥160.830B	Baseline comparison
HPC conservative	100 GPU/node; 40%×¥400/h	-15.71%	▲¥159.767B	▲¥156.351B	Not sufficient
Displayed HPC base	150 GPU/node; 55%×¥600/h	-12.13%	▲¥147.256B	▲¥147.160B	Improves; no payback
HPC high	200 GPU/node; 70%×¥900/h	-6.07%	▲¥118.914B	▲¥126.359B	Still insufficient
NPV@8% break-even	+¥43.337B/y in 2030–2035	8.00%	¥0.000B	▲¥33.927B	Beyond HPC alone

Nominal Cumulative FCF and Payback Extension			
Period	Year / cumulative FCF	Year / cumulative FCF	Year / cumulative FCF
2025–2027	2025 / ▲¥40.000B	2026 / ▲¥86.000B	2027 / ▲¥136.000B
2028–2030	2028 / ▲¥178.000B	2029 / ▲¥210.000B	2030 / ▲¥224.946B
2031–2033	2031 / ▲¥227.053B	2032 / ▲¥220.143B	2033 / ▲¥202.944B
2034–2036	2034 / ▲¥174.064B	2035 / ▲¥133.489B	2036 / ▲¥92.915B
2037–2039	2037 / ▲¥52.341B	2038 / ▲¥11.767B	2039 / ¥28.807B PAYBACK
Extension basis	2036–2039	FCF ¥40.574B/year	No renewal CAPEX/tax/WC

Cash-Flow Completeness and Production-Case Inputs			
Completeness item	Current treatment	Required production-case addition	Financial effect
Working capital	0	Receivables/payables/inventory timing by year	FCF / payback
Cash tax	0	Loss carry-forward, tax rate and timing	FCF / NPV
Financing	Excluded	Debt/equity, interest, fees and repayment	Equity IRR / DSCR
Residual / salvage	Excluded	Terminal asset value by asset class	Terminal FCF / NPV
Maintenance / renewal	Not separately shown	Overhaul, replacement and lifecycle cadence	OPEX / CAPEX
Node/HPC cash flow	Not separately shown	Revenue/OPEX/CAPEX/tax/LH2 cold energy by node	Node/HPC payback / ROIC

Financial Interpretation & KPI Linkage

The displayed case adds a separately priced distributed-HPC service without counting internal mobility compute as revenue. 2035 HPC revenue is ¥15.610B and incremental HPC EBITDA is ¥12.730B. Annual project FCF is positive from 2032, but cumulative nominal FCF remains ▲¥133.489B at the end of 2035. Nominal and discounted payback are not achieved; project IRR is -12.13% and NPV at 8% is ▲¥147.256B.

The largest unresolved drivers are contracted HPC capacity, paid GPU utilization, accelerator-hour price, node delivery cost, fleet OPEX, hydrogen volume/cost, platform pricing, depreciation-heavy investment, working capital, tax, and renewal CAPEX. KPI improvements affect finance only through explicit volume, price, cost, and investment equations.

Node Deployment Plan

Node Deployment Milestone	Cumulative nodes	Objective
By 2030	4	Initial AVP / MaaS validation
By 2032 (scenario)	12	Fleet synchronization
By 2035	36	City-scale operation

KPI and Financial Equation

KPI	Financial impact	Required equation
Utilization	Revenue increase	available vehicle-hours × monetized trips/hour
Fleet idle ratio	OPEX/CAPEX efficiency	idle hours × avoidable fleet cost
Matching rate	Trips increase	requests × matching × dispatch × realization
Dispatch success	Revenue stability	matched trips × successful dispatch
Energy cost/trip	Margin	energy input × unit cost ÷ trips
Node/HPC utilization	CAPEX efficiency	paid GPU-hours ÷ available external GPU-hours
GPU-hour price	Compute revenue	contracted HPC revenue ÷ paid GPU-hours
Trips/day	Revenue scale	fleet × available hours × trips/hour
Load factor	Revenue efficiency	occupied passenger-km ÷ available seat-km
EBITDA margin	Cash generation	EBITDA ÷ revenue
Operating margin	Accounting profit	operating profit ÷ revenue

Screening-Case Scope Check

Completion Check	Status
P&L / cash-flow asset schedules reconciled	Production-case reconciliation required
H2 internal/external cohorts non-overlapping	Production-case cohort split required
Depreciation rows re-summed	YES
IRR recalculated from displayed FCF	YES
NPV at 8% calculated	YES
Nominal project payback calculated	YES
Discounted payback calculated	YES
Node payback separately calculable	Node-level cash flow required
Rendered Word/PDF layout verified	YES
Working capital included	Excluded from screening case
Cash taxes included	Excluded from screening case
Financing structure included	Excluded from screening case
Residual value included	Excluded from screening case
Renewal CAPEX included	Not separately modeled
Node/HPC revenue/OPEX/CAPEX split	Not separately modeled

Investment Decision Logic

- Autonomous-mobility profitability is dominated by utilization, load, price, energy, fleet OPEX, and infrastructure CAPEX - not labor reduction alone.
- Node deployment may convert parking assets into operational infrastructure only after contract, COGS, utilization, permit, and residual-value assumptions are defined.
- AVP-based infrastructure sharing can support private users and MaaS fleets, but revenue cannot be recognized twice across base fares, platform fees, node fees, and settlements.
- Hydrogen-energy integration can reduce OPEX only if metered end-to-end efficiency, production cost, storage, losses, and uptime support the claimed unit cost.
- Vehicle OEMs, mobility-platform providers, operators..., suppliers, and finance entities require contractually explicit authority and accountability; the responsibility structure is proposed, not current fact.
- Distributed-HPC sales improve project returns but do not achieve nominal payback by 2035; contracted capacity, staged investment, lower CAPEX/OPEX, or an alternative financing structure remains necessary.

Financial Driver Targets

Driver	2030 target	2035 target	Financial impact	Evidence status
Utilization	0.75	0.85	primary volume driver	target
Fleet idle ratio	0.25	0.15	OPEX/CAPEX efficiency	target
Energy cost/trip	≤¥500	≤¥350	margin	target
HPC price/GPUh	¥700	¥600	compute revenue	scenario
Matching rate	≥90%	≥95%	demand realization	target
HPC paid utilization	35%	55%	compute revenue	scenario
Trips/day	73	140	revenue scale	scenario
Load factor	0.70	0.80	revenue efficiency	scenario
EBITDA margin	42.4%	35.8%	cash generation	calculated
Operating margin	-55.7%	-27.7%	accounting profit	calculated

Investment Assessment

Investment assessment	Target	Displayed case	Status
Nominal project payback	By 2035	Not achieved	BELOW HURDLE
2035 annual FCF	>0	¥40.574B	POSITIVE
IRR	≥12%	-12.13%	BELOW HURDLE
NPV @8%	≥0	▲¥147.256B	BELOW HURDLE
Discounted payback	By 2035	Not achieved	BELOW HURDLE

Distributed-HPC sales improve 2035 annual FCF to ¥40.574B, but cumulative investment is not recovered by 2035; contracted compute revenue and CAPEX/OPEX optimization remain necessary to meet the 8% NPV and 12% IRR hurdles.

Design Constraints & System Physics - Source Register

Values are classified as Official value, Technical target, Scenario assumption, or Reference only. A source link does not convert a scenario assumption into a fact. Rows with no page-specific or value-specific support are retained as references but must not be used as validated inputs.

Category	Variable	Corrected value / status	Source	Evidence classification / use
Constraint	Insurance requirement reference	Defined only after exact provision identified	MLIT guidance	Reference only
	Operational-time regulation	Define by service/vehicle/operator law	MLIT transport	Reference only
	Safety-margin requirement	Safety-case derived; no generic numeric value	MLIT safety	Reference only
	Traffic regulation	Applicable rules must be mapped to ODD	NPA self-driving	Official framework
	Weather constraint	ODD-specific threshold required	MLIT AD	Reference only
Cost	Charging infrastructure cost	10–50M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	EV cost benchmark	5–10M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	Hydrogen-station CAPEX	150–300M JPY retained as scenario range	IEA Future of H2	Scenario assumption
	Software cost ratio	10–30% retained as scenario range	OECD finance	Scenario assumption; not source-specific
Demand	Reference-market population	Define after the target geography is selected	Statistics Bureau	Official table; value not repeated here
	Japan population	122.68M at 2026-08-01	STAT_POP_202608	Official value
	Taxi fare	Local fare table required; no national 2,000–2,500 constant	MLIT taxi	Official framework
	Trips per person	2.3/day retained only as survey assumption	MLIT transport survey	Scenario assumption
	Target-region vehicle stock	Use latest Chubu table	Chubu vehicle stats	Official table; validate date
Energy	Battery storage cost trend	Declining; exact year/system boundary required	NREL battery	Reference only
	Electricity intraday price	13.0 JPY/kWh retained only with period definition	METI ELEC p14	Scenario/period-specific
	Electricity retail price	30.2 JPY/kWh	METI ELEC p14	Official cited benchmark
	Electricity spot price	12.24 JPY/kWh retained only with period definition	METI ELEC p14	Period-specific
	Hydrogen LHV	33.3 kWh/kg	DOE H2 LHV	Physical constant
	Hydrogen pathway roundtrip efficiency	Lower than direct electricity; system boundary required	NREL H2 pathway	Reference only
	Hydrogen production cost	150–200 JPY/kg retained as scenario	IEA Future of H2	Scenario assumption
	Hydrogen sales price	300–400 JPY/kg retained as scenario	NEDO H2 library	Scenario assumption
	Hydrogen specific energy	120 MJ/kg	DOE H2 LHV	Physical constant
	Hydrogen storage cost	DOE 2020 target: \$333/kg-H2 stored	DOE 2020 STORAGE	Technical target; not achieved current value
	Hydrogen storage gravimetric	DOE 2020 target: 4.5 wt%	DOE 2020 STORAGE	Technical target; not achieved current value
	PEMFC direct-H2 efficiency	~60% LHV	DOE PEMFC 60	Technology comparison; not 22%
	LNG energy density	13.9 kWh/kg retained	IEA Future of H2	Reference; page/definition required
	LNG volumetric energy	20.8 MJ/L retained	IEA Future of H2	Reference; page/definition required
	Liquid-H2 volumetric energy	~8 MJ/L retained	DOE H2 storage	Reference
	Generation cost benchmark	LCOE reference only	IEA generation cost	Reference only
	PV high-output benchmark	>1,100kWh/kW/yr	NEDO PV p14	Benchmark
	PV output benchmark	1,000–1,100kWh/kW/yr	NEDO PV p14	Benchmark
Financial	SOFC efficiency	~60% scenario benchmark	DOE fuel cells	Technology-dependent
	Discount rate	8–12% retained as screening range	OECD finance	Scenario assumption
	Infrastructure return	Project-specific	JBIC	Reference only
	Investment hurdle	Project-specific	OECD finance	Reference only
Infra	Project IRR benchmark	10–15% retained as hurdle range	JBIC	Scenario assumption
	H100 cloud price benchmark	Public cloud ≈\$11/GPUh; model ¥400–900/h	Google A3	Benchmark; model is scenario
	GPU/system power	H100 up to 700W; model 1.2kW/GPU + PUE 1.15	NVIDIA H100	Official GPU limit + scenario
	Hydrogen infrastructure	Definition/reference only	IEA Future of H2	Reference only
Logistics	Standard freight fare	Distance/time and vehicle-class tables; no national 52 JPY/kg constant	MLIT STANDARD FARE	Official framework
Market	Global EV sales share	15% retained only with year/definition	IEA EV Outlook	Period-specific
	Heavy-truck new sales (Japan)	73k retained only after exact JAMA table validation	JAMA statistics	Validate year/category
	Japan EV sales share	1% retained only with year/definition	IEA EV Outlook	Period-specific
	Total new vehicle sales (Japan)	4.421M retained only after exact JAMA table validation	JAMA statistics	Validate year/category
Policy	Japan 2035 electrified-vehicle target	100% new passenger-car sales; scope definition required	Public policy context	Policy definition required
	Electrified-vehicle definition	BEV/PHEV/HEV/FCEV	METI energy policy	Policy definition
	Japan powertrain distribution	Use a dated JAMA powertrain table	JAMA statistics	Reference only
Regulation	Accident response obligation	Required	NPA L4	Official
	Administrative cancellation	Applicable	NPA L4	Official
	Autonomous-driving subsidy availability	Program-specific	MLIT subsidy	Verify FY/eligible costs
	Current safety-policy baseline	Applicable	MLIT safety	Official framework
	Government safety policy	Defined	MLIT AD	Official framework
	Human-operated-vigilance requirement	Depends on operation type	NPA L4	Official framework
	Infrastructure regulation	Defined by installed equipment/site	MLIT AD	Map exact law
	Insurance requirement	Required; exact coverage required	MLIT AD	Map exact provision
	International standards relevance	Relevant, not automatically legally binding	MLIT safety	Reference
	Legal-liability framework	Defined across product/operator/road law	MLIT safety	Map exact provisions
	Level 4 summary availability	Available	NPA L4	Official
	Local-government consultation	Required where specified	NPA L4	Official
	Monitoring requirement	Required where specified	NPA L4	Official
	Non-specific operation treatment	Defined	NPA interpretation	Official interpretation
	ODD requirement	Required	NPA L4	Official
	Operation plan submission	Required	NPA portal	Official
	Passenger transport rule	Applicable by service model	e-Gov Passenger Rule	Official law
	Permit cancellation	Applicable	NPA L4	Official
	Police reporting	Required where specified	NPA L4	Official
	Public-road operation framework	Defined	MLIT AD	Official
	Remote monitoring	Allowed under defined conditions	NPA L4	Official
	Road Traffic Act	Applicable	e-Gov Road Traffic Act	Official law
	Road Transport Vehicle Act	Applicable	e-Gov Vehicle Act	Official law
	Road-use regulation	Defined	MLIT safety	Official
	Running-environment condition	Required	MLIT AD	Official
	Specific autonomous-operation permit	Required	NPA portal	Official
	Subsidy eligible cost	Program-specific	MLIT subsidy	Verify current notice
	Suspension authority	Applicable	NPA L4	Official
	Transport safety law reference	Defined	MLIT safety	Official
	Vehicle Act enforcement order	Applicable	e-Gov order	Official law

Note on Information Sources and Intellectual Property

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